

# VALD



## Practitioner's **Intermediate** Guide to **Force Plates**

A comprehensive guide for practitioners taking the next step in integrating force plates into practice.

# Some Words from the Force Plate Experts



**Alex Natera**

Performance Science Manager,  
New South Wales Institute of Sport

[in](#) [X](#) [f](#) [@alex.natera](#)

“ Force plates are an essential element in establishing **effective biomechanical deterministic models** for sport performance.

Once I understand the **underpinning neuromuscular physical qualities** that “predict” performance, I can then strategize my training interventions to **improve the physical quality** and movement outcome.

**By longitudinally tracking force plate results, I can manipulate my training plan** and determine whether my focus should target the further development of an underpinning physical quality or whether it should focus more on motor skill development for a given sporting task. ”



**Matt Jordan**

Player Health &  
Performance Consultant,  
Jordan Strength

[in](#) [X](#) [@jordanstrength](#)

“ Using force plates allows us to take a more targeted approach, identifying where athletes may be lacking in strength capacity from a neuromuscular standpoint.

**This not only helps improve their overall preparation but also addresses the long and often difficult recovery journey following injuries like ACL tears.**

It's important to recognize that regaining full performance can take many months, if not years. Our work highlights how mechanical variables, such as stretch-shortening cycle function, can remain impaired for extended periods.

Weekly jump tests on force plates give us a valuable tool to model recovery, track progress, refine training and most importantly, **identify whether an athlete is moving towards their goals or veering off course.** ”



**Dr. Daniel Cohen**

Co-Founder,  
ForceDecks

[in](#) [X](#) [@dr\\_danielcohen](#)

“ It's exciting to see how sports science, performance, and medical practitioners leverage force plate insights to answer performance and injury-related questions while fostering more thoughtful inquiries.

The ability of time-constrained force and duration metrics to distinguish between elite and sub-elite athletes and **uncover elite neuromuscular qualities** highlights the value of comprehensive CMJ analysis.

**These metrics are also highly sensitive to load response, providing actionable insights for monitoring and training adaptations.**

As qualitative and quantitative analysis of CMJ waveforms evolves, similar detailed kinetic analysis is being applied to other jump types, uncovering metrics that further illuminate neuromuscular and musculotendinous qualities.

At the same time, practitioner-researchers are addressing data management challenges by systematically reducing and refining metrics for practical application. ”



## Lachlan James

Associate Professor, Course Coordinator: Master of Strength and Conditioning,  
**La Trobe University**

[in](#) [X](#) [v](#) [@doclachjames](#)

“ The goal of strength-power assessment is to isolate and contextualize the different aspects of maximal force expression, and force platforms are an essential part of this process.

A general model can include a countermovement jump, drop jump and isometric mid-thigh pull, with a peak measure and a time-based measure extracted from each of these tests.

Along with lifting strength (e.g., a 1RM), this will provide a **comprehensive overview of an athlete's force production capabilities where unique information is maximized and redundant information is reduced.** ”



## Dan Howells

Founder & Lead Mentor,  
**Collaborate Sports**

[in](#) [X](#) [v](#) [@collaborate\\_sports](#)

“ Advancements in force plate technology have **revolutionized how practitioners design programs** for athletes and clients. By accelerating the assessment process, practitioners gain access to a more comprehensive set of solutions, allowing for deeper insights into program design and enhancing performance outcomes.

**The automation provided by VALD technology streamlines the analysis phase**, freeing up valuable time for practitioners to focus on interpreting results and applying these insights to the next stages of planning and training. ”



## Duncan French

Senior Vice President of Performance,  
**UFC Performance Institute**

[in](#) [X](#) [v](#) [@dr\\_duncan\\_french](#)

“ Performance reflects an athlete's neuromuscular potential, and optimizing it begins with evaluating effectiveness.

Force plates play a vital role in assessing abilities like generating, dissipating, or resisting force — skills critical to athletic success.

**By simplifying complex data collection and analysis, they allow practitioners to focus on enhancing key qualities.**

This precision ensures that training interventions are tailored to the factors most crucial for achieving performance outcomes. ”



## Jo Clubb

Founder and Sports Science Consultant  
**Global Performance Insights**

[in](#) [X](#) [v](#) [@joclubbsportsci](#)

“ Force plates **empower science** and medical practitioners with objective data that was previously inaccessible.

By analyzing force-time curves, time-point and phase metrics and asymmetries, we gain an intricate understanding of how athletes generate and transfer force.

This granular view of movement **allows for the fine-tuning of training and rehabilitation** plans to develop physical capacities and address specific concerns.

Exploring this deeper level of analysis not only strengthens data-informed decision-making but also **supports individualized, targeted interventions that enhance physical development, injury prevention and rehabilitation strategies.** ”

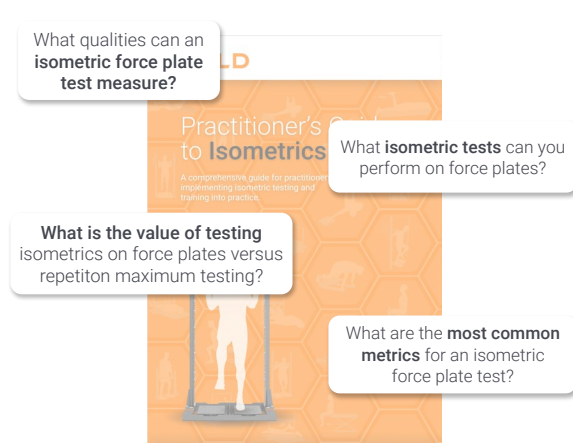
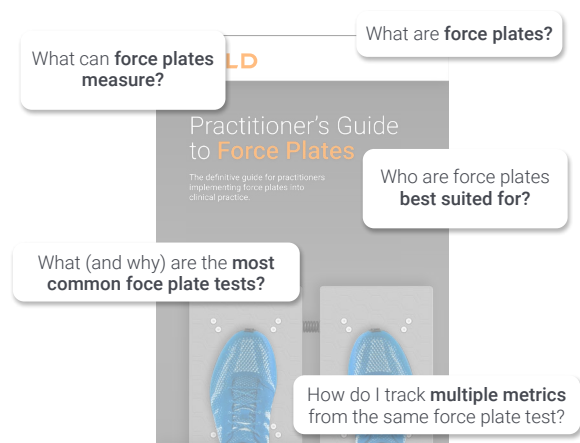
# What Will You **Not** Get from This Guide?

This guide is for healthcare and sports performance practitioners who have already experienced the power and benefits of force plate technology for themselves.

Rather than repeating the basics, we assume you are already familiar with the fundamentals of force plates, have performed a range of the main assessment types (Jumps, Isometrics, Balance and Functional) and are now looking for the next-level use cases for this technology.

To help you deepen your knowledge and refine your skills, you should be familiar with a few key concepts about force plates before getting too far into this guide:

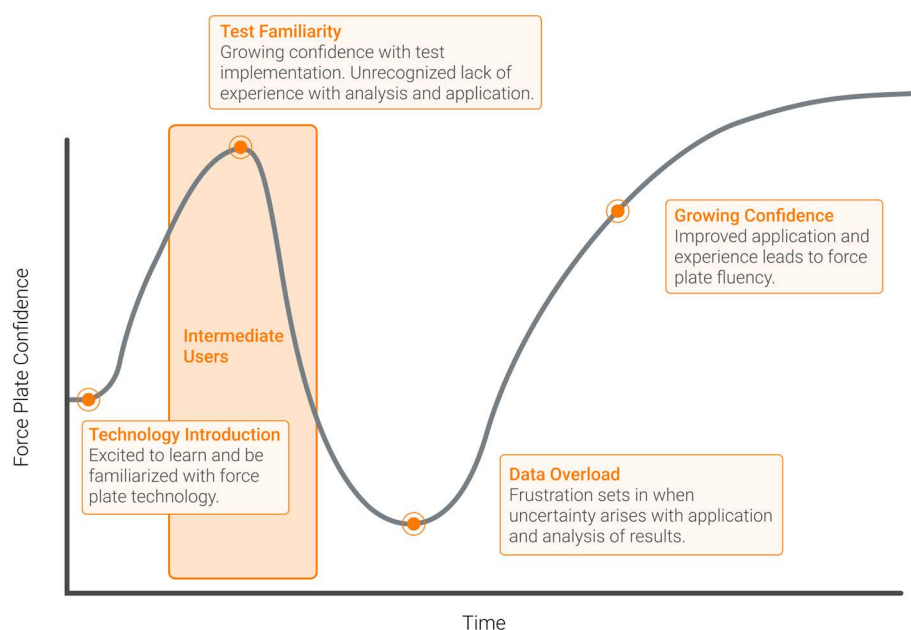
- What force plates are (and what they measure)
- Who they are best suited to
- Which are the most common tests (and why)
- The additional insights force plates provide over traditional methods of measurement
- How to track and assess different metrics from the same force plate test



If you feel uncertain about any of these areas, we recommend exploring VALD's other Practitioner's Guides, such as the Practitioner's Guide to Force Plates and the Practitioner's Guide to Isometrics. These resources provide essential background knowledge that will enhance your understanding as you move into the more complex topics covered in this guide.

...beware of overconfidence in force plate testing and **embrace the opportunity to continue your learning journey.**

For users who have reached this intermediate stage of force plate knowledge and experience, congratulations! However, the road to force plate mastery is a long one, so beware of overconfidence in force plate testing and embrace the opportunity to continue your learning journey.



A typical user's journey as they learn and apply force plates in practice.

## What **Will** You Get from This Guide?

This guide provides insights into force plate technology, enabling you to optimize testing, understand key metrics and tailor assessments to your athletes' or patients' needs. The following chapters will explore the **process** of force plate test selection, metric appraisal and compound assessment to best apply force plate knowledge in practice.

By taking a systematic approach to force plate assessment, this guide will help you make personalized, effective and efficient testing sessions a staple of your practice. Force plate technology should seamlessly fit within your processes, expanding the work you do across a range of performance and rehab populations.

Get ready to accelerate your force plate competency. **Let's jump in.**

# Contents

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In this document, we will **cover a wide range of topics and answer some common questions**, including:

**Tip:** Click any heading to jump to section

Where Do  
**Force Plates Fit?**

**Selecting** Force Plate  
Assessments

Force Plate  
**Assessments**

Concepts in **Force  
Plate Interventions**

Developing **Strength**

Advanced **Metrics,  
Indexes** and **Profiles**

Developing  
**Reactive Strength**

Developing  
**Explosiveness**

Force Trace  
**Analysis**

# Introduction



## Blind Spots

It is common for practitioners with a year or less of force plate experience to be competent (and confident), yet still lack experience in many scenarios and applications.

Keep an eye out for these cards throughout this guide, which will highlight common blind spots for intermediate users to watch out for.

## Fundamentals of Force Plate Metrics

Force plates measure two simple things: force and time. All other force plate metrics are either calculated or derived from them.

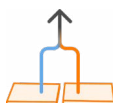
For those starting out, **focusing on measured metrics** like peak force, contraction time and time to peak force is a reliable, straightforward approach.

For those starting out, focusing on **measured** metrics like peak force, contraction time and time to peak force is a reliable, straightforward approach. This helps streamline decision-making and avoid distractions, especially in fast-paced environments like team sports or healthcare.

### Total Vertical Force

Left Plate Vertical Force + Right Plate  
Vertical Force = **Total Force**

With dual force plate systems, total force is calculated by **summing the forces from both force plates**.



### Vertical Force

**Force** = Mass x Acceleration

Force is the **product of mass and acceleration** and its measurement is underpinned by Newton's second law of motion. In the context of force plate testing, mass is typically represented by the athlete's body weight (BW) while standing still.



Beyond these **measured** metrics are more **derived** metrics. However, these require a deeper understanding of physics, performance and technology, as derived metrics applied without a full understanding of their benefits and shortcomings can lead to misguided performance and rehabilitation decisions.

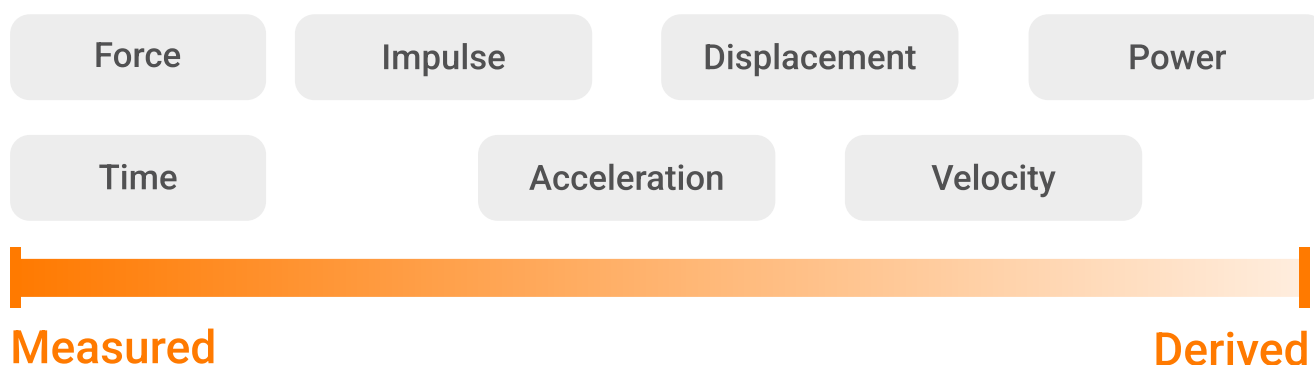
### Expert Tip:

Reliability is a key measurement characteristic, but metric selection shouldn't rely solely on an arbitrary 10% threshold for "good" variability. A metric's value also depends on its responsiveness to inputs like loading or injury. Ultimately, the balance between reliability and responsiveness – its "signal-to-noise ratio" – determines its ability to detect meaningful change.



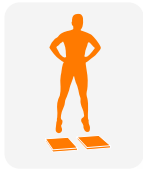
**Daniel Cohen, PhD**  
Co-Founder of ForceDecks

## Force Derivative **Continuum**



Jumping into complex data without a solid foundation can lead to confusion and suboptimal outcomes. There are three key reasons for this potential pitfall:

1. Practitioners often have **limited information about their client or team** before the first force plate assessment. By refining the metric selection process, you provide stability and consistency in the measurements that influence decisions with your patients and athletes.
2. Human measurements are inherently complex. Therefore, **adding metrics into a complex system creates noise** that can overwhelm even experienced professionals if they do not truly understand the metric's definition. Elite practitioners keep key performance goals in focus with basic measurements, solving complex problems with simple solutions.
3. When **multiple measurement options exist for a single quality** (e.g., jump height), using a stable measurement strategy like the [impulse-momentum \(IMP-MOM\)](#) calculation will yield more accurate and reliable results, provided the jump is executed to an appropriate level.



**Many metrics** exist for a **single quality...**  
Even for the same test.

| Force Expression               | Power Expression             | RFD Expression             |
|--------------------------------|------------------------------|----------------------------|
| Peak Takeoff Force             | Peak Power / BM              | Eccentric Deceleration RFD |
| Force at Zero Velocity         | Vertical Velocity at Takeoff | RSI-Mod                    |
| Eccentric Peak Force           | Concentric Mean Power        | Concentric RFD             |
| Eccentric Deceleration Impulse | Eccentric Peak Power         | Concentric RPD             |

#### Key

**RSI-Mod:** Reactive Strength Index-Modified  
**BM:** Body Mass  
**RFD:** Rate of Force Development  
**RPD:** Rate of Power Development

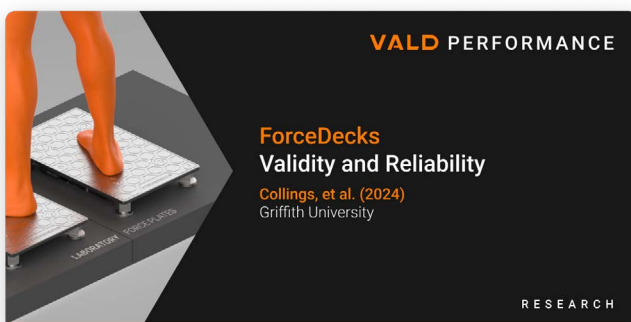
Process-based metric selection enables practitioners to focus on accurate metrics that represent meaningful performance and rehab outcomes.

## Force Plate **Essentials**

Without a solid foundation in force plate testing methods and protocols, **even the most advanced technology can produce inaccurate results.**

Force plates, as a technology, simply measure the forces encountered during the tasks performed on them. Therefore, validity (the ability to accurately measure a quality of interest) and reliability (the consistency of measurement accuracy) are heavily influenced by human input.

These influences include practitioner setup procedures, testing cues and task effort and execution by the patient or athlete.



#### Expert Tip:

Weighing phases are incredibly important for your athletes. During the weighing phase, ensure that there are no distractions (e.g., nearby conversations) to ensure the athlete is still, without sudden movements or gestures. This ensures a standardized and repeatable setup process.



**Ryan McLaughlin**  
Sport Scientist, VALD BDM

**VALD**

# Understand the Test

## Expert Tip:

Metric reliability is essential for confidently interpreting observed changes. Reliability depends on factors such as population, experience with the assessment, proper cueing, motivation and engagement.

Since reliability is cohort-specific, practitioners should track day-to-day reliability in their setting, noting that it can improve over time.



**Daniel Cohen, PhD**  
Co-Founder of ForceDecks

Appropriate test selection is essential for quality force plate assessments and requires a clear understanding of both the test and its relevance to the goals of the performance or clinical team.

For example, elite sprint performance relies on rapid force development and coordination at high speeds.

## Rate of Force Development (RFD)

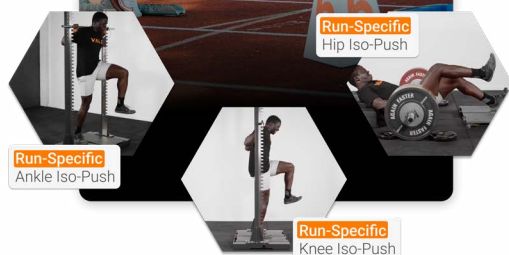
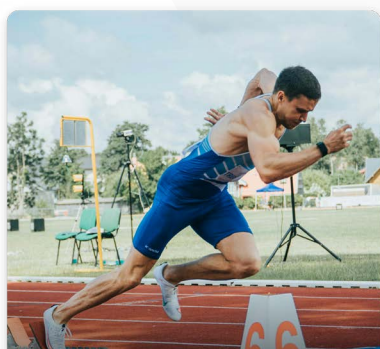
$$\frac{(\text{Force at Time Point 1} - \text{Force at Time Point 2})}{(\text{Difference Between Time Points 1 and 2})} = \text{RFD}$$

RFD can be visually illustrated as the slope of a force trace at a given time period. RFD provides insights into an athlete's capacity for force development and explosive strength.



By understanding the task, practitioners can **cut redundancy and sharpen the accuracy** of their testing insights.

To assess strength in this context, the [Run-Specific Isometric Strength Assessment \(RSISA\)](#) popularized by [Alex Natera](#) can be used to measure RFD and peak force in a unilateral, running-specific position. This provides precise insights into sprint-specific characteristics that traditional methods, such as a 1-repetition maximum (1RM) test, may be too generalized to detect.



By understanding the task, practitioners can focus on the key performance qualities that matter, cutting redundancy and sharpening the accuracy of testing insights.

## Expert Tip:

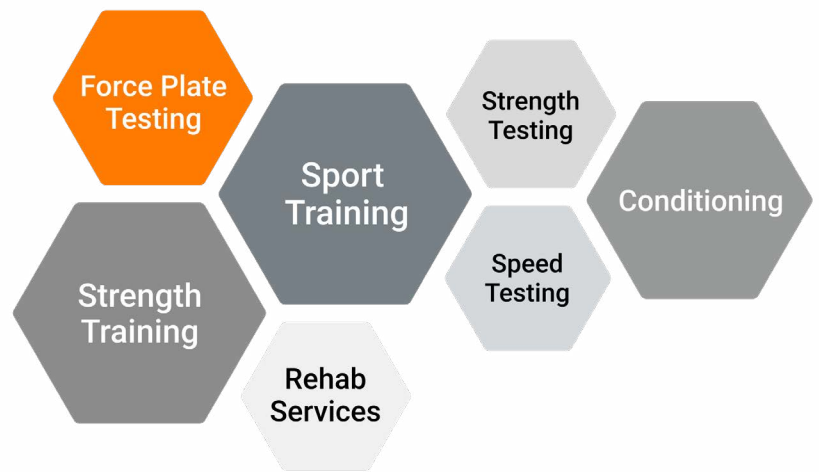
Don't judge an assessment as "good" or "bad" based purely on how it replicates the **movement** of your chosen sport or activity. Instead, focus on implementing tests that replicate the **loads** and **demands** of the sport or activity.



**Gabriel Manoel**  
Physical Therapist, VALD CSM

**VALD**

# Where Do Force Plates Fit?



## Performance Outcomes

Outcomes drive every practitioner's program, whether they are helping athletes jump higher, accelerate faster, change direction more efficiently or restore function and reduce pain.

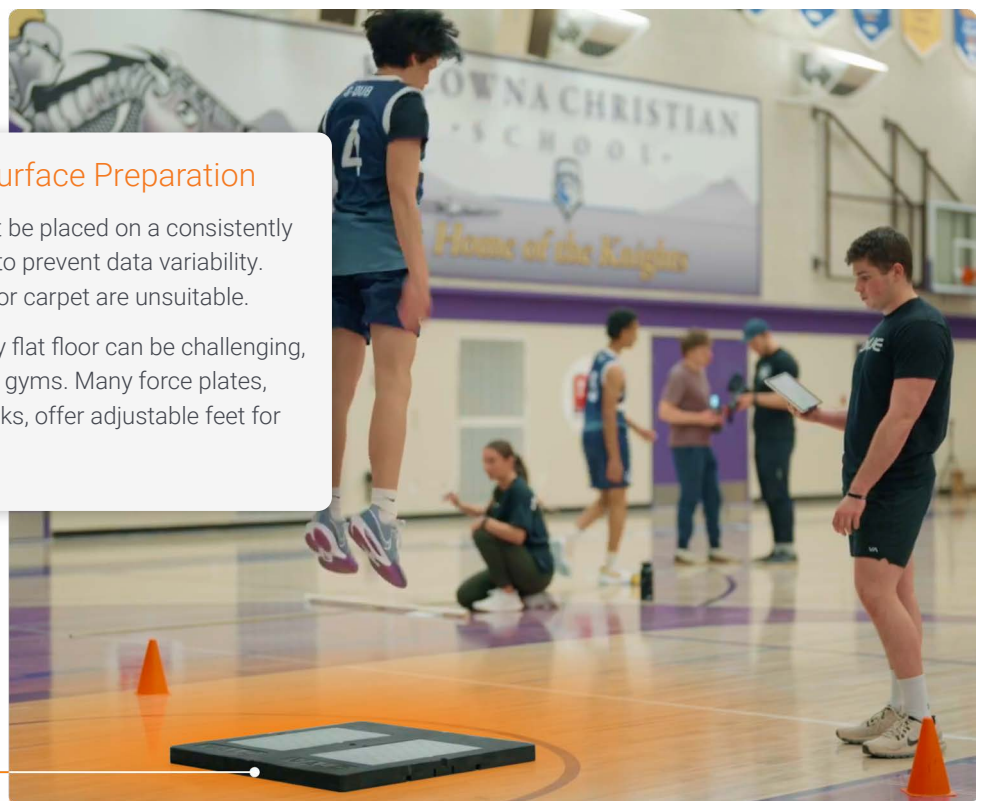
Force plate metrics allow functional and neuromuscular changes to be quantified, influencing rehab and performance outcomes.



### Blind Spot: Surface Preparation

Force plates must be placed on a consistently flat, hard surface to prevent data variability. Surfaces like turf or carpet are unsuitable.

Finding a perfectly flat floor can be challenging, especially in older gyms. Many force plates, such as ForceDecks, offer adjustable feet for precise leveling.



# Force plates...offer a precise window to assess the **physical qualities that impact performance outcomes...**

Measuring performance outcomes (e.g., a patient's mobility or athlete's fitness level) directly is challenging and often unreliable due to the complex interaction of factors that influence day-to-day tasks and athletic performance.

Force plates, however, allow for a controlled testing environment and offer a precise window to assess the physical qualities that impact performance outcomes, helping practitioners address key limitations affecting performance.

## Identifying **Rehab Trends**

Like performance, a rehabilitation provider's practice is driven by quality outcomes, whether working to restore normal walking patterns, improve balance and stability, reduce pain or enhance functional strength.

Force plate metrics provide critical insights into a patient's recovery status and their response to reconditioning or programming throughout their rehabilitation journey.

Subjective reports, like pain-free squatting or improved daily task performance, are valuable for guiding rehabilitation and building patient connections.

However, force plate assessments provide objective data, allowing practitioners to visualize precise measurements and trends, which offers greater confidence in assessing patient progress and outcomes.

Force plates create a standardized testing environment that precisely evaluates key neuromuscular and biomechanical attributes, guiding clinicians to tailor interventions and address impairments efficiently.

This precision enables practitioners to track meaningful changes, assess whether the program is driving desired adaptations and make informed adjustments.

Objective data enhances treatment individualization, optimizing the patient's recovery process.

## Questions to **Ask Before Testing**

When conducting force plate tests, several key considerations must be kept in mind to ensure your data is reliable, valid and meaningful. These considerations revolve around the athlete's experience with the test, the context in which the test is conducted, how cues are given and best practices for data collection.

| Consideration      | Question                                                     | Reasoning                                                                                                   | Resolution                                                                                              |
|--------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Familiarity</b> | Is this the first time they have performed the test?         | Test familiarity can drastically affect test performance.                                                   | Training exposure to testing positions and intensities can improve athlete familiarization.             |
| <b>Outcome</b>     | What is the primary objective of your testing outcome?       | Monitoring and performance testing have different goals and are, therefore, conducted differently.          | Align the assessment conditions to the goal of the tests (e.g., recovered versus non-recovered).        |
| <b>Cueing</b>      | How are you cueing your patients or athletes during testing? | Test metrics may change based on the cues provided.                                                         | Provide standardized cues that are consistent with the quality you are assessing.                       |
| <b>Volume</b>      | How many repetitions should be performed for each test?      | Test reliability can change if metrics are averaged over multiple repetitions or taken from a single trial. | Performing 3-5 repetitions per test allows practitioners to average the data for more reliable results. |

# Selecting Force Plate Assessments

Many elite practitioners have a structured performance process that allows them to easily understand and interpret athlete insights. These processes aim to encompass all athletic performance.

Force plates will usually have a place in **multiple levels of the assessment hierarchy, yet are simply a tool that serves the overall goal.**

In this section, Dr. Alex Natera shares the assessment hierarchy he has followed across a successful career of almost 25 years in elite sports. While exact implementation is likely to differ, this hierarchy can be applied throughout performance, health and return to play scenarios to best meet the needs of the athlete and interdisciplinary team.

## Expert Tip:

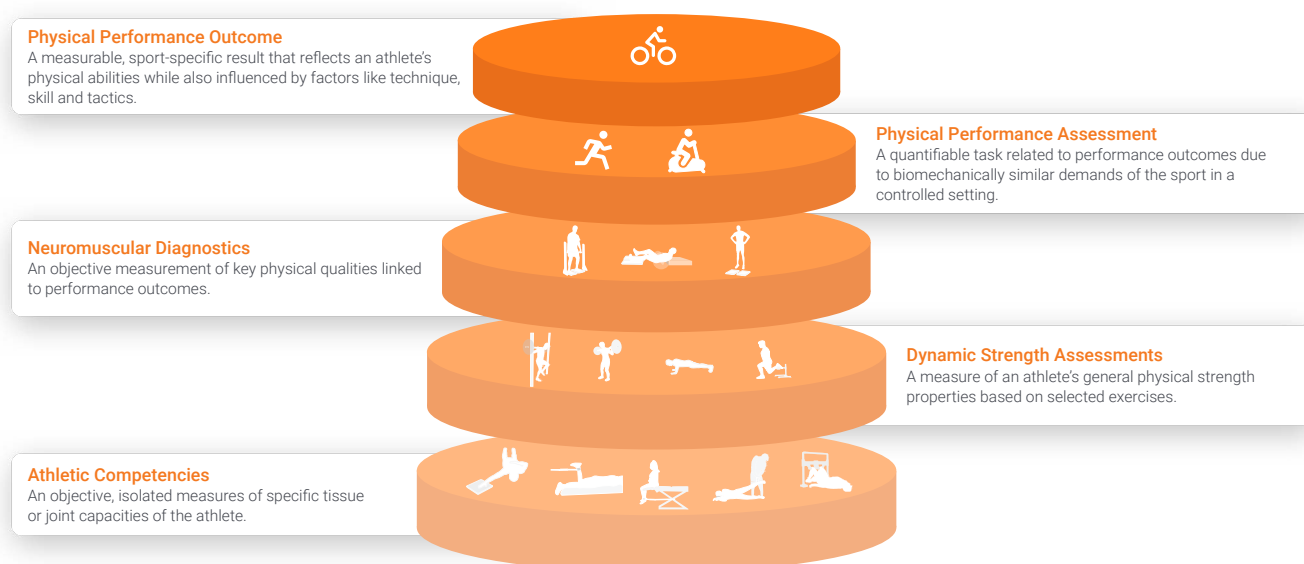
Developing assessment frameworks and hierarchies is an essential learning process for practitioners to develop more informed decisions around training interventions and training emphases required for their clients.



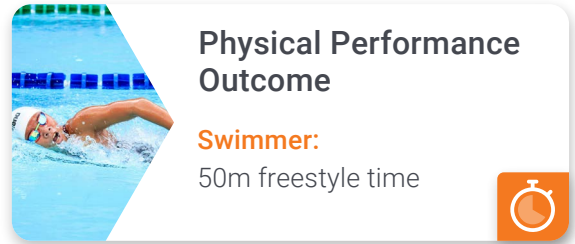
**Alex Natera, PhD, MSc**  
Head of Sport Science at NSWIS



Each “layer” of the hierarchy serves the next, culminating in (typically) a single physical performance outcome. Force plates will usually have a place in multiple levels of the assessment hierarchy, yet are simply a tool that serves the overall goal.



# Physical Performance Outcome



**Physical performance outcomes should be quantifiable performance attributes that influence the outcome of the competition...**

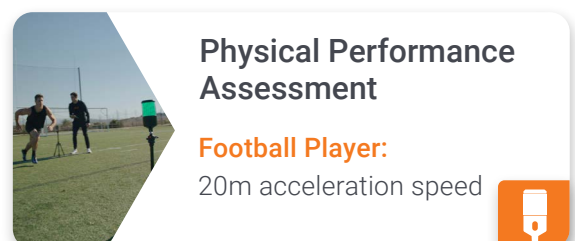
Physical performance outcomes should be quantifiable performance attributes that influence the outcome of the competition or reveal aspects of the athlete's capabilities, such as:

- 50m freestyle time for a swimmer;
- 2,000m time for a rower;
- Total weight (snatch + clean and jerk) for an Olympic weightlifter; or
- Post-contact meters gained or percentage tackle competition for a football player.

However, physical performance outcomes are not attributed solely to physical attributes. Factors that are more difficult to quantify – like technique, skill, tactics, “fitness” and arousal – also come into play.

These are important to consider in athlete performance, especially when making decisions. While improvements can always be made, they are typically outside the realm of the practitioner and are usually associated with tactical decisions or the sport coaches.

# Physical Performance Assessment



The physical performance assessment aims to quantify variables that are closely tied to physical performance outcomes in a controlled, repeatable environment.

For example, practitioners are often interested in outputs such as cycle ergometer peak power output for sprint cyclists or peak velocity in a 100m sprinter. These tests of athlete performance represent the physical performance assessment due to their repeatability and strong carryover to sport.

### Vertical Velocity

$$\text{Velocity} = \frac{\text{Net Force} \times \text{Time}}{\text{BM}}$$

Vertical velocity is calculated by multiplying the net force (the force produced greater than the subject's mass) by time to get the **net impulse**. Then, the net impulse is divided by the athlete's BM to determine velocity.



### Displacement

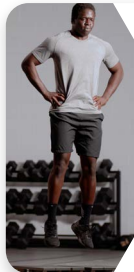
$$\text{Displacement} = \text{Average Velocity} \times \text{Time}$$

Displacement is another complex calculation that can be simplified by averaging an athlete's vertical velocity within a given time period and multiplying it by the time elapsed.



If a top performer in a physical performance assessment does not also rank as a top performer in the physical performance outcome, one may assume that there may be a tactical or technical deficiency, informing the coaching team's programming interventions.

## Neuromuscular Diagnostics



### Neuromuscular Diagnostics

#### Basketball Player:

Reactive strength index (RSI)



Neuromuscular diagnostic assessments involve assessing physical qualities that may influence the physical performance outcome.

There may be one key physical quality that explains a large variation of the physical performance outcome or a small range of physical qualities that contribute to it.

Neuromuscular diagnostic assessments should be established using sensitive testing equipment like force plates and employ reliable and valid metrics.

Depending on the physical performance outcome, the neuromuscular diagnostic assessment, for example, may be as simple as utilizing the **countermovement jump (CMJ)** to measure lower body power, the **isometric mid-thigh pull (IMTP)** to measure maximal, absolute or relative force output or the **drop jump (DJ)** to assess reactive strength.

If the improvements in identified deficits do not lead to an enhancement in the physical performance assessment, there may be a delayed transfer.



This delay occurs as the athlete learns to effectively apply the new physical quality (e.g., peak power) to their sport. Alternatively, the multi-disciplinary team may need to implement motor skill and coordination interventions or specific training modalities to facilitate a more direct transfer from the neuromuscular quality to the physical performance assessment.

### Power

$\text{Force} \times \text{Velocity} = \text{Power}$



As the product of force and velocity (also known as the rate of work), power can only be measured during dynamic activities. Power can be presented as an absolute metric in watts (W) or relative to the athlete's BM in kilograms (W/kg). Jump tests are common assessments of power due to their versatility in speed and ability to load.

Conversely, it is an equally important problem if the physical performance assessment improves with no accompanying change in the neuromuscular diagnostic.

This may be indicative of a metric selection issue. Selecting the “right” metric to most likely represent a change in the physical performance outcome is integral to the assessment process (see [“Understand the Test”](#) for more information).

## Dynamic Strength Assessments



### Dynamic Strength Assessment

Rugby Player:

3 RM deadlift



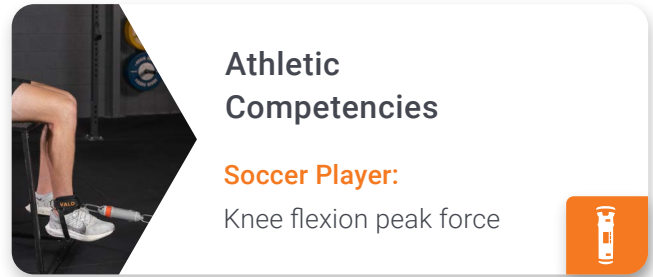
Dynamic strength assessments identify an athlete's ability to move loads and serve as the basis for prescribing interventions to promote improvements in various neuromuscular diagnostic assessments.

Although dynamic strength assessments (e.g., back squat or power clean) are underpinned by physical qualities, they are also heavily influenced by training age, anthropometry and technical mastery. Particularly where heavy loads are concerned, psychological aspects related to arousal and confidence also play a role.

Dynamic strength assessments...**are also heavily influenced by training age, anthropometry and technical mastery.**

Dynamic strength assessments can be classified as general exercises. As such, they lack the mechanical specificity of the physical performance outcome and the physical performance assessment. Therefore, practitioners must always contextualize the results of dynamic strength assessments.

# Athletic Competencies



Athletic competencies, like a hip hinge pattern within a movement screen or isolated muscle function and capacity assessments (e.g., a heel raise capacity test), form the base of the assessment hierarchy.

...isolate the specific function as much as possible to give the practitioner **greater insights into precise limitations and deficiencies.**

It is important to note that athletic competency assessments can be maximal assessments, such as a maximal effort Nordic hamstring curl assessment on [NordBord](#) or an isometric knee flexion test using a dynamometer such as [DynaMo](#) or [ForceFrame](#). The key is that the assessment should isolate the specific function as much as possible to give the practitioner greater insights into precise limitations and deficiencies.

Athletic competency assessments provide a more controlled environment to highlight interlimb asymmetries, postural control, range of motion (ROM) and muscular imbalance issues. They can also be used to highlight weak links within the kinetic chain that otherwise could be hidden within more dynamic assessments.

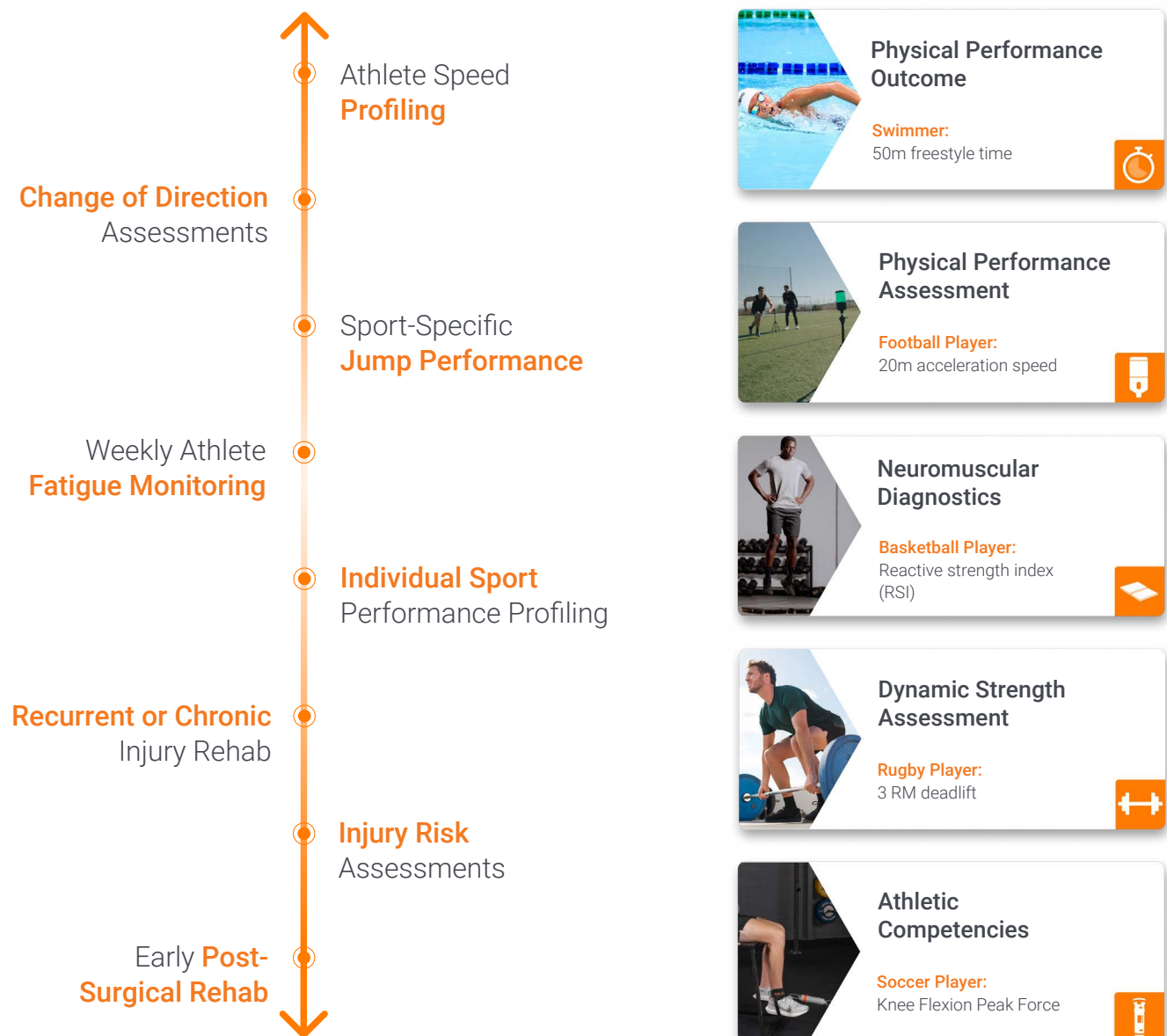


# Assessment Hierarchy Applications

Alex Natera, PhD, MSc  
Head of Sport Science for NSWIS

## Top-Down

Assessing competencies that most closely represent the physical performance outcome first, ensuring athlete training performance closely corresponds with sporting performance.



## Bottom-Up

Addressing individual impairments (e.g., joint ROM and isolated force production) to ensure a healthy and safe return to more dynamic and open tasks.

# Force Plate **Assessments**

While a range of common force plate tests were introduced in the [Practitioner's Guide to Force Plates](#), each test type has nuances that must be understood to ensure your data is as accurate as possible.

In this section, we revisit the wide range of tests that force plates can perform, this time focusing on more advanced concepts, protocols and applications that are important to understand for the intermediate user.

## **Isometric** Tests

Isometric testing provides a precise, objective assessment of an athlete's maximal force capacity and RFD from a given position without requiring dynamic, movement-based testing.

By measuring the peak and time-based force metrics in controlled positions, practitioners gain insights into an athlete's neuromuscular capacity and strength characteristics, which are extremely valuable for performance profiling, monitoring and injury risk assessment.

The key benefits of isometric testing compared to traditional, dynamic effort tests include its safety (no movement means reduced injury risk during testing), repeatability and ability to provide detailed force metrics. When performed on force plates, the data obtained provides even greater insights into neuromuscular performance.

Isometric testing and training have seen a recent uptick in popularity due to their high level of transfer to rehab and performance, ease of setup and the amount of data they provide a practitioner. For an in-depth explanation of isometric testing and training, check out the [Practitioner's Guide to Isometrics](#).



### **Blind Spot: Pretension**

Explain the pretension phase to the athlete to ensure they perform it accurately. If the athlete does not initiate their maximal effort contraction from an appropriate level of pretension, it may compromise the validity of their test data.

## Isometric Mid-Thigh Pull (**IMTP**)

### **1. Test Description**

A multi-joint isometric exercise to assess total body strength and force production capacity, IMTP is one of the most used force plate tests in a range of environments.

It is designed to replicate the second pull phase of the Olympic clean exercise and involves the participant pulling a fixed bar and pushing their feet into the force plates maximally.

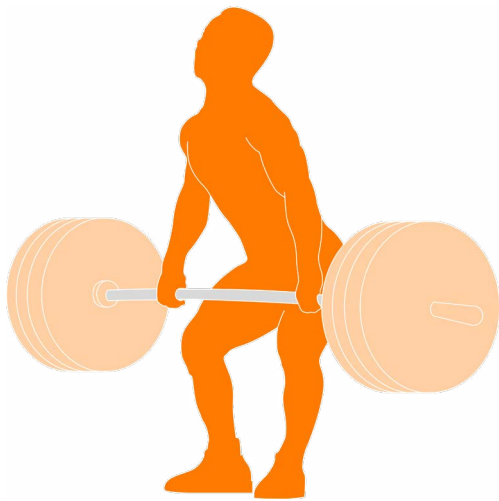


Image depicting the second pull of the Olympic clean exercise, a position that is highly transferrable to data received from an IMTP.

## 2. Applications

The IMTP measures peak force during an isometric contraction, providing a safe and controlled assessment of maximal force capacity. Its simplicity and low injury risk make it ideal for in-season use, where minimizing fatigue is critical.

The test tracks strength adaptations over time, compares force outputs, and complements other strength and power measures. These measures help coaches make informed decisions and monitor progress, even for athletes with high training loads.

By testing specific strength under high loads, the IMTP identifies gaps in resilience and ensures athletes across sports, from rugby to golf, have the strength needed for competition.

### Normative Data General Population: Men

|                                      | Age Group          | 25 <sup>th</sup> Percentile | 50 <sup>th</sup> Percentile | 90 <sup>th</sup> Percentile |
|--------------------------------------|--------------------|-----------------------------|-----------------------------|-----------------------------|
| <b>Peak Force</b><br><br><b>IMTP</b> | 31-40<br>Years old | 2233N                       | 2655N                       | 3619N                       |
|                                      | 41-50<br>Years old | 2102N                       | 2459N                       | 3371N                       |
|                                      | 51-60<br>Years old | 2095N                       | 2370N                       | 2992N                       |
|                                      | 61-70<br>Years old | 1835N                       | 2140N                       | 2759N                       |
|                                      | >70<br>Years old   | 1626N                       | 1880N                       | 2342N                       |

### Normative Data General Population: Women

|                                      | Age Group          | 25 <sup>th</sup> Percentile | 50 <sup>th</sup> Percentile | 90 <sup>th</sup> Percentile |
|--------------------------------------|--------------------|-----------------------------|-----------------------------|-----------------------------|
| <b>Peak Force</b><br><br><b>IMTP</b> | 31-40<br>Years old | 1453N                       | 1738N                       | 2545N                       |
|                                      | 41-50<br>Years old | 1404N                       | 1637N                       | 2422N                       |
|                                      | 51-60<br>Years old | 1353N                       | 1543N                       | 2231N                       |
|                                      | 61-70<br>Years old | 1241N                       | 1440N                       | 1784N                       |
|                                      | >70<br>Years old   | 1123N                       | 1267N                       | 1641N                       |

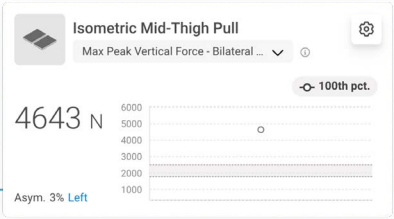
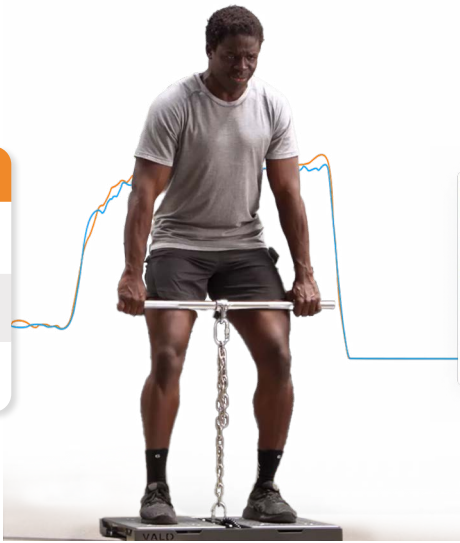
### 3. Test Insights

An individual's peak force production ability and, where relevant, RFD qualities.

### 4. Common Test Metrics

- **Peak vertical force (PVF) (N):** The maximum force applied onto the force plates during the test. This can be presented absolutely (N) or relative to BM (N/kg).
- **RFD at various time points (100ms, 150ms, 200ms, etc.):** The rate of isometric force production between specific time points.
- **PVF asymmetry (%):** The percent deficit of peak force production from the weaker limb when compared to the stronger limb.

| Metric                      | Insight                                                                         |
|-----------------------------|---------------------------------------------------------------------------------|
| PVF                         | Common indicator of maximal total body extensor strength.                       |
| RFD<br>(50ms, 100ms, 200ms) | Indicator of explosive strength qualities between various time points.          |
| PVF Asymmetry               | Common indicator of interlimb differences in lower extremity strength capacity. |



### 5. Practical Considerations

In the IMTP, the expression of an individual's lower body strength may be limited by their grip capacity, obscuring their true maximal force. To avoid this, hand wraps or tape can be used to fix the hands to the bar, reducing grip demand during the test. For this reason, some users may opt for the Isometric Squat instead.

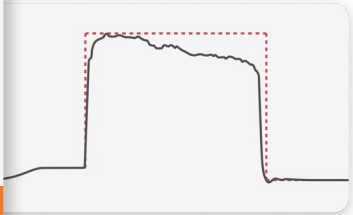
For valid, accurate and reliable data, consider the individual's exposure to similar movements (e.g., Olympic lift derivatives or barbell deadlifts) and intensities (e.g., push isometrics) before testing. With proper coaching cues, the IMTP can provide useful baseline strength numbers across populations.

#### Expert Tip:

Show the individual the live force trace during the test and cue them to “draw a box.” Connecting their effort and strategy to live output with this cue – to produce a steep vertical rise in force – typically results in improved reliability and magnitude of RFD metrics, particularly in early exposure to the test.



Daniel Cohen, PhD  
Co-Founder of ForceDecks



# Isometric Squat (**Iso Squat**)

## 1. Test Description

A multi-joint isometric exercise to assess lower body strength and force production qualities. The test is performed in the quarter squat position, with the individual pushing maximally into the force plates from under a fixed bar.

Unlike the IMTP, the Iso Squat does not rely on grip strength or straps, though shoulder or neck discomfort may be a limiting factor for some individuals and can negatively affect their results.

## 2. Applications

By holding a static squat position while maintaining contact with a fixed bar, individuals can exert maximal force safely, enabling the measurement of key metrics like peak force, RFD and interlimb symmetry. This provides a clear assessment of an athlete's maximal strength and can reveal asymmetries or force deficits not always evident in dynamic tests.

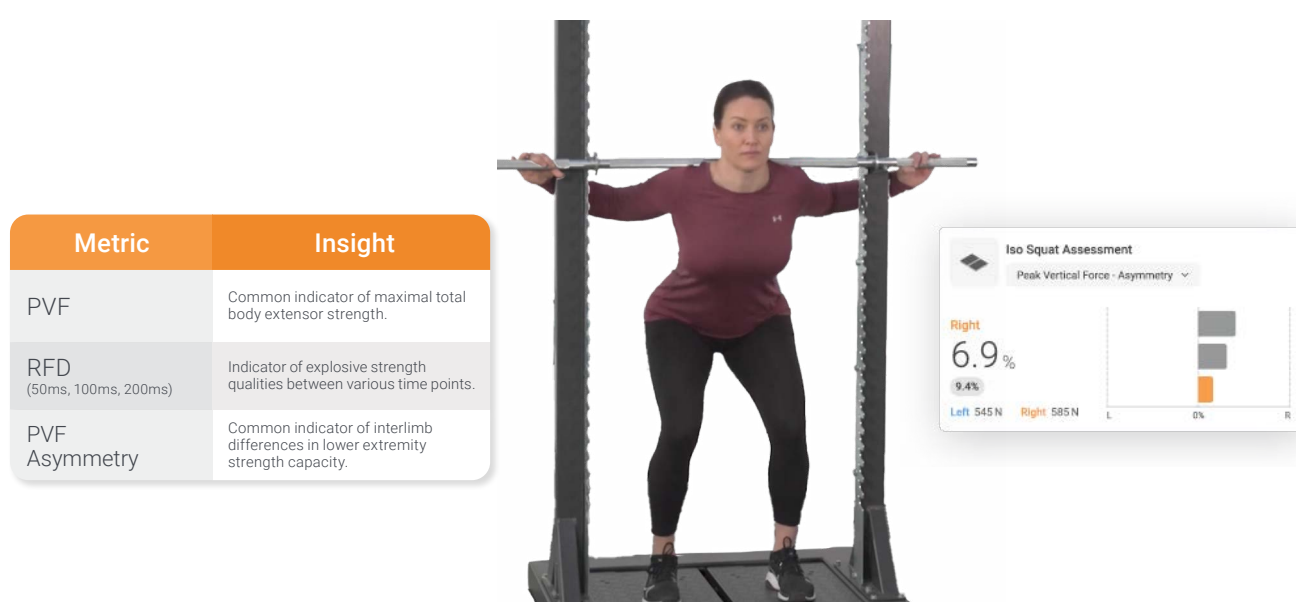
The isometric nature of this test, like the IMTP, makes it ideal for regular in-season monitoring, helping to minimize fatigue and reduce injury risk.

## 3. Test Insights

Peak force production and maximal force-generating capabilities in a vertical extension strategy without influence from secondary strength qualities (e.g., grip strength).

## 4. Common Test Metrics

- **PVF (N):** The maximum force applied onto the force plates during the test. This can be presented absolutely (N) or relative to BM (N/kg).
- **RFD at various time points (100ms, 150ms, 200ms, etc.):** The speed at which force is produced isometrically, measured at discrete time points (epochs).
- **PVF (%):** The percent deficit of peak force production from the weaker limb when compared to the stronger limb.



## 5. Practical Considerations

Individual performance in the Iso Squat and IMTP may vary depending on the athlete's biomechanics and training history. Therefore, it is important that practitioners properly consider their rationale for selecting the Iso Squat or IMTP.

The Iso Squat is ideal for testing RFD, as it involves simply pushing downward with the upper back compressed against a fixed barbell. In contrast, the IMTP can result in “force leakage,” as individuals must focus on both pushing into the plates and pulling the bar to maximize force production.

For individuals with less developed upper back muscles, isometric belt squats are a viable option to mitigate potential discomfort in that area when performing a maximal effort Iso Squat.

Like the IMTP, the Iso Squat offers sport-specific insights as a lower body strength test. It's particularly useful for field and court sports, where running and jumping place high demands on the lower extremities.

For niche sports like alpine skiing, which require both load tolerance and endurance, modified tests such as a maximal belt squat push for time can help assess peak force endurance.



# Athletic Shoulder (ASH) Test Battery

## 1. Test Description

This test battery isolates the function of the shoulder girdle through long lever force assessments in a prone position. Providing insight into peak force production and RFD qualities in three common sporting positions (I, Y, T), the ASH testing series uncovers force capacities of the shoulder that other shoulder assessments cannot.

The ASH Test can be performed with a number of technologies, including force plates, [handheld dynamometers](#) or fixed-frame dynamometers.

## 2. Applications

The ASH Test battery is a valuable tool for assessing shoulder strength, stability and endurance, especially for individuals in overhead sports or recovering from upper extremity injury. It provides objective measures of shoulder force output in various positions, offering insights into scapular control and overall stability.

Using force plates or dynamometers, practitioners can quantify strength and identify asymmetries that may increase injury risk. The ASH battery helps detect deficits and guides targeted programming to enhance shoulder stability and endurance. For more information on the ASH test and other upper extremity assessments, download the [Practitioner's Guide to Shoulders](#) today.

## 3. Test Insights

The ASH Test battery reveals an individual's peak force production ability, RFD qualities and any interlimb asymmetries. Particularly useful in overhead athletes who throw, swing, punch or experience heavy impact forces through the upper extremity.

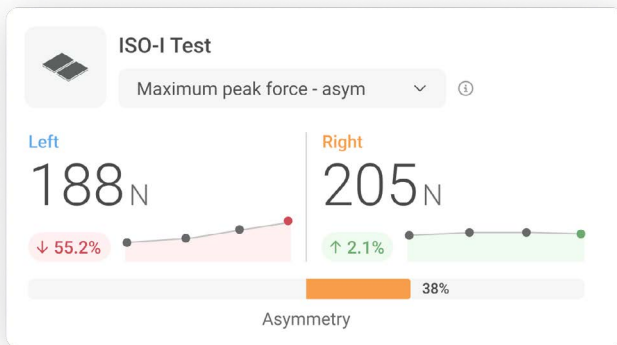
## 4. Test Positions

- **I: Prone**, 180° of shoulder abduction, head supported on a surface of the same height as the force plates, opposite arm resting at the subject's side.
- **Y: Prone**, 135° of shoulder abduction, head supported on a surface of the same height as the force plates, opposite arm resting behind the subject's back.
- **T: Prone**, 90° of shoulder abduction, head supported on a surface of the same height as the force plates, opposite arm resting behind the subject's back.



## 5. Common Test Metrics

- **Peak force:** The maximum force imparted onto the force plates during the test.
- **RFD (100ms):** The speed at which force is generated, measured from the start of movement until the first 100ms of the effort.
- **Asymmetry (%):** The percent deficit of peak force production from the weaker limb when compared to the stronger limb.



| Metric                      | Insight                                                                |
|-----------------------------|------------------------------------------------------------------------|
| Peak Force                  | Common indicator of maximal shoulder girdle strength.                  |
| RFD<br>(50ms, 100ms, 200ms) | Indicator of explosive strength qualities between various time points. |
| Asymmetry %                 | Common indicator of strength capacity.                                 |

## 6. Practical Considerations

For all ASH Test battery variations, a standardized approach is essential. Many practitioners use tape on the floor to mark the desired angles, making it easier to guide patients and athletes on shoulder positioning and ensure repeatability.

Ensuring consistent hand placement during each test (either centered on the plate or off to one side) minimizes changes in system weight readings.

Individuals resting their forearms on the force plates may produce different readings compared to those who simply place their hands. Using a foam pad or yoga block to support the head also helps isolate the shoulder joint fully.

# Run-Specific Isometric Strength Assessments (RSISA)

## 1. Test Description

The RSISA series assesses the kinetic capabilities of the ankle, knee and hip. The tests are conducted in supine or standing, encouraging maximum effort in positions that mimic the kinetic demands of high-speed running (see each assessment below).

## 2. Test Insights

The RSISA test provides practitioners with insights into both the peak force and rapid-generating capacity of the lower extremity. Each assessment is unilateral, providing asymmetry data representative of unilateral capacity, rather than bilateral execution strategy. Force output data can be compared to population norms to inform training interventions.

## 3. Test Positions

- **Ankle Iso-Push:** Standing on a single leg with a slight forward trunk lean, hip in 0° of extension, knee fully extended and the ankle begins in 0-5° of plantarflexion. The forefoot of the testing limb is directly underneath the bar and the contralateral hip is flexed to 90°.
- **Knee Iso-Push:** Standing on a single leg with a vertical torso, hip and knee flexed to 35-45° (135-145° angle) with the foot flat on the ground. The midfoot of the testing limb is directly underneath the bar and the contralateral leg is held in a comfortable position for the subject.
- **Hip Iso-Push:** The individual is lying supine with their mid-back (just below the shoulder blade) supported on a 15cm (6in) box or firm cushion. Knee angle between 35-45° of knee flexion (135-145° angle at the knee) and the hip to be flexed between 15-25° (hip-trunk angle to be at 155-165°). A heavy barbell or fixed implement is resting in the hip crease, with a pad recommended for subject comfort.



Ankle Iso-Push



Knee Iso-Push



Hip Iso-Push

### 3. Common Test Metrics

- **PVF:** The maximum force imparted onto the force plates during the test.
- **Relative peak force:** The maximum force imparted onto the force plates during the test, divided by the patient or athlete’s BM.
- **Force at 100ms:** The maximum amount of force produced 100ms after the start of the test was registered.

| Metric              | Insight                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------|
| PVF                 | Common indicator of maximal total body strength. Can also be used to calculate asymmetry. |
| Force at 100ms      | Indicator of explosive strength qualities in short time periods.                          |
| Relative Peak Force | Common indicator of force production relative to the individual’s BM.                     |

### 4. Practical Considerations

At maximal sprinting velocity, the body experiences and produces ground reaction forces of 4-5 times BW ([Udofa, 2017](#)), highlighting the importance of assessing isometric strength in these running-specific positions.

In assessments where multiple joint angles and positions must be controlled for, it is best to measure hip, knee and ankle angles once pretension has been established, as small variations in joint angle can influence performance outcomes significantly.

These assessments aid in targeted programming and can inform both load and intensity prescription for overcoming, yielding and quasi-isometric training.

In rehabilitation and return-to-play settings, peak forces remain important. However, RFD becomes especially relevant as rapid force generation is required during return to play.

# Jumps

## Countermovement Jump (CMJ)

### 1. Test Description

At this stage, readers will all be aware of what the CMJ is and what it tests. However, the CMJ yields more metrics than any other force plate test. Understanding the best ways to apply it and how to best interpret its results is beyond the scope of any beginner's resource.

### 2. Applications

The CMJ is a widely used tool to assess lower body power, neuromuscular fatigue and readiness. Key metrics like jump height, peak concentric power and concentric impulse provide insights into performance and load response.

Strategy-based metrics, such as contraction time and eccentric deceleration duration, reveal movement execution details. Tracking these metrics over time helps evaluate athlete performance relative to sport-specific standards and monitor fatigue or training adaptations.

With its simplicity, repeatability, and low burden, the CMJ is ideal for regular monitoring, aiding in training adjustments during in-season and off-season periods. In rehabilitation, it tracks progress by measuring performance metrics like jump height and compensatory strategies through force plates, ensuring balanced reconditioning.

### 3. Test Insights

Maximal power output, jumping performance and jumping strategy can be assessed through jump height, along with detailed insights into vertical deceleration and acceleration qualities.

Normative Data General Population: Men

| <div>Jump Height</div> <br>CMJ | Age Group          | 25 <sup>th</sup> Percentile |        | 50 <sup>th</sup> Percentile |        | 90 <sup>th</sup> Percentile |        |
|-------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------|--------|-----------------------------|--------|-----------------------------|--------|
|                                                                                                                   | >18<br>Years old   | 27.9cm                      | 10.9in | 33.4cm                      | 13.1in | 45.1cm                      | 17.8in |
|                                                                                                                   | 18-25<br>Years old | 31.8cm                      | 12.5in | 37.1cm                      | 14.6in | 48.1cm                      | 18.9in |
|                                                                                                                   | 26-30<br>Years old | 29.8cm                      | 11.7in | 35.0cm                      | 13.8in | 46.0cm                      | 18.1in |
|                                                                                                                   | 31-35<br>Years old | 27.4cm                      | 10.8in | 32.6cm                      | 12.8in | 44.0cm                      | 17.3in |
|                                                                                                                   | 36-40<br>Years old | 24.9cm                      | 9.8in  | 29.7cm                      | 11.7in | 41.4cm                      | 16.3in |
|                                                                                                                   | 41-50<br>Years old | 22.4cm                      | 8.8in  | 26.7cm                      | 10.5in | 37.0cm                      | 14.6in |

## Normative Data General Population: Women

| Age Group                                                                                                      | 25 <sup>th</sup> Percentile |                    | 50 <sup>th</sup> Percentile |                    | 90 <sup>th</sup> Percentile |                                          |
|----------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------|-----------------------------|--------------------|-----------------------------|------------------------------------------|
| <b>Jump Height</b><br><br>CMJ | >18<br>Years old            | 21.1 <sub>cm</sub> | 8.3 <sub>in</sub>           | 24.7 <sub>cm</sub> | 9.7 <sub>in</sub>           | 33.2 <sub>cm</sub><br>13.0 <sub>in</sub> |
|                                                                                                                | 18-25<br>Years old          | 21.6 <sub>cm</sub> | 8.5 <sub>in</sub>           | 25.9 <sub>cm</sub> | 10.2 <sub>in</sub>          | 25.9 <sub>cm</sub><br>10.2 <sub>in</sub> |
|                                                                                                                | 26-30<br>Years old          | 20.4 <sub>cm</sub> | 8.0 <sub>in</sub>           | 24.9 <sub>cm</sub> | 9.8 <sub>in</sub>           | 35.6 <sub>cm</sub><br>14.0 <sub>in</sub> |
|                                                                                                                | 31-35<br>Years old          | 17.9 <sub>cm</sub> | 7.0 <sub>in</sub>           | 21.9 <sub>cm</sub> | 8.6 <sub>in</sub>           | 34.7 <sub>cm</sub><br>13.6 <sub>in</sub> |
|                                                                                                                | 36-40<br>Years old          | 16.7 <sub>cm</sub> | 6.6 <sub>in</sub>           | 20.3 <sub>cm</sub> | 8.0 <sub>in</sub>           | 31.7 <sub>cm</sub><br>12.5 <sub>in</sub> |
|                                                                                                                | 41-50<br>Years old          | 14.8 <sub>cm</sub> | 5.8 <sub>in</sub>           | 18.1 <sub>cm</sub> | 7.1 <sub>in</sub>           | 26.1 <sub>cm</sub><br>10.3 <sub>in</sub> |

## 4. Common Test Metrics and Applications

Broadly, the CMJ is used to assess both healthy and rehabilitating individuals in profiling and load response monitoring (LRM) applications.

**Profiling** aims to quantify the individual's neuromuscular status at a given time point and typically involves comparison to peer groups and relevant norms. Assessments are periodic (separated by weeks or months) and are ideally performed in a recovered state.

**Monitoring**, in contrast, denotes more frequent assessments (i.e., 1-2/week) and focuses on determining individual responses to loading and is often performed in unrecovered states (i.e., 24-72 hours after a competitive game or demanding training session).

The following table from Dr. Daniel Cohen summarizes some features of data collection according to the individual's condition (healthy vs. injured/rehabilitation) and the application (profiling vs. LRM).

|         | Profiling                                                                                                                                          | LRM<br>Short-term   Long-term                                                                                                                                                                                                   |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Healthy | Comparison of athlete against norms for sport, position, level and age.                                                                            | <b>Quantifying individual response</b><br><b>Short Term:</b> Residual fatigue/recovery status following competition or intense training.<br><b>Long Term:</b> Response to a loading program, period of training or competition. |
| Rehab   | Comparison of patient status to pre-injury profile (if available), peer healthy data and rehab stage reference data for asymmetry and performance. | <b>Short Term:</b> Injured vs. healthy limb response to intense session or series of sessions.<br><b>Long Term:</b> Adaptations in response to period of rehabilitation.                                                        |

Below, we list some of the available CMJ metrics that provide neuromuscular insights and indicate whether they are prioritized in the contexts described by Cohen's framework.

| Key                                       |                  | PROFILING |   | MONITORING |   |
|-------------------------------------------|------------------|-----------|---|------------|---|
| ✓ Healthy                                 | ✓ Rehabilitation |           |   |            |   |
| Metric                                    |                  | H         | R | H          | R |
| Jump Height                               |                  | ✓         | ✓ | ✓          | ✓ |
| FT:CT or RSI-Mod                          |                  | ✓         |   | ✓          |   |
| CMJ Stiffness                             |                  | ✓         |   | ✓          |   |
| <b>ECENTRIC</b>                           |                  |           |   |            |   |
| Eccentric Deceleration Phase Duration     |                  |           |   | ✓          |   |
| Eccentric Mean Power                      |                  | ✓         | ✓ | ✓          | ✓ |
| Eccentric Peak Velocity*                  |                  | ✓         | ✓ | ✓          | ✓ |
| Countermovement Depth**                   |                  | ✓         | ✓ | ✓          | ✓ |
| Force at Zero Velocity                    |                  |           |   | ✓          |   |
| Eccentric Deceleration RFD / BM           |                  | ✓         |   |            |   |
| Eccentric Deceleration RFD Asymmetry      |                  | ✓         | ✓ |            | ✓ |
| Eccentric Deceleration RFD (Left & Right) |                  |           |   |            | ✓ |
| Eccentric Deceleration Impulse            |                  | ✓         | ✓ |            |   |
| Eccentric Deceleration Impulse Asymmetry  |                  | ✓         | ✓ |            | ✓ |
| <b>CONCENTRIC</b>                         |                  |           |   |            |   |
| Peak Power / BM                           |                  | ✓         | ✓ |            |   |
| Concentric Mean Force or Impulse          |                  | ✓         |   | ✓          | ✓ |
| Concentric Mean Force (Left & Right)      |                  |           |   |            | ✓ |
| Concentric Impulse – 100ms                |                  | ✓         | ✓ | ✓          | ✓ |
| Concentric Impulse – 100ms (Left & Right) |                  |           |   |            | ✓ |
| Force at Peak Power Asymmetry             |                  |           | ✓ |            | ✓ |
| Force at Peak Power (Left & Right)        |                  |           |   |            | ✓ |
| P2 Concentric Impulse Asymmetry           |                  |           | ✓ |            | ✓ |
| <b>LANDING</b>                            |                  |           |   |            |   |
| Peak Landing Force                        |                  | ✓         |   |            | ✓ |
| Peak Landing Force Asymmetry              |                  | ✓         | ✓ |            | ✓ |
| Peak Landing Force (Left & Right)         |                  |           |   |            | ✓ |

\* **Eccentric Peak Velocity** is used as a technique check – to verify that the countermovement is being performed at a sufficient velocity and with a level of consistency that gives them confidence in the values achieved for other eccentric metrics.

\*\* **Countermovement Depth** changes during rehabilitation need to be interpreted with a different lens than in the healthy individual.

An increase is considered a negative (fatigue) response in a healthy individual, while during rehabilitation, it may represent a willingness to go into deeper knee flexion (a positive adaptation in an athlete with a knee injury who has had a restricted range). This also applies to metrics strongly associated with it (i.e., phase durations, FT:CT or RSI-Mod), hence their lower priority in rehabilitation.



### Impulse

$$\text{Force} \times \Delta \text{Time} = \text{Impulse}$$

Impulse is the product of force and time and is visualized by calculating the area underneath the force-time curve. Different measures of impulse have strong relationships with key outcomes such as jump height, making it an important metric to understand.

## 5. Practical Considerations

For healthy individuals, [Cohen et al. \(2020\)](#) recommend an eccentric peak velocity of at least -1.2m/s to ensure a valid jump. Slower eccentric speeds may not accurately reflect the individual's deceleration strategy.

While jump height, concentric/eccentric impulses, and peak power are also commonly analyzed, [research shows](#) that specific metrics like FT:CT can be sensitive markers of fatigue.

### Hot Topic: RSI-Mod or FT:CT?

When assessing fatigue status with the CMJ, some coaches prefer RSI-Mod, while others prefer FT:CT.



However, both measure similar aspects of reactive strength and jump efficiency by creating a ratio of the time spent contracting (pre-jump) and the time spent in the air.

[Eben et al. \(2010\)](#) suggested the use of RSI-Mod, and Cormack and colleagues' work has validated the use of FT:CT. Both metrics underpin similar qualities; therefore, emphasizing high jumps with short time on the ground will elicit optimal number outputs.

# Squat Jump (SJ)

## 1. Test Description

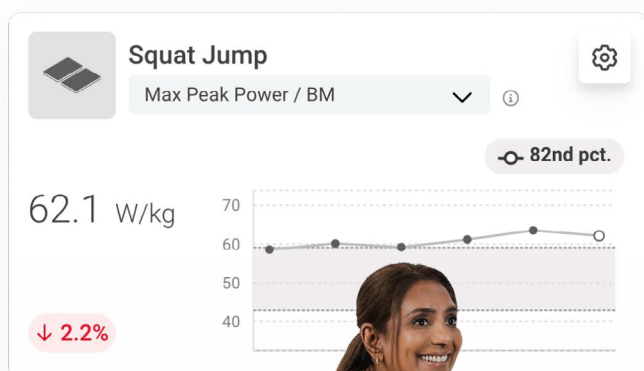
Similar to the CMJ, the SJ is a measure of lower body explosive power. The key difference is that the SJ is purely concentric and eliminates the eccentric utilization of the CMJ.

## 2. Test Insights

The SJ provides insights into the ability to generate maximal concentric power from a static start. Performance characteristics like jump height can be compared to the CMJ to calculate an athlete's eccentric utilization ratio (EUR).

## 3. Common Test Metrics

- **Jump height (IMP-MOM):** The maximum height (vertical displacement) jumped in a test, measured from net concentric impulse, determining takeoff velocity.
- **Peak power:** Power is calculated as force x velocity and provides an insight into the explosiveness of the individual from a concentric-only strategy.
- **Concentric impulse asymmetry:** Percentage increase of the stronger limb, compared to the weaker limb, when measuring the summed total force (force x time) produced in the concentric phase.



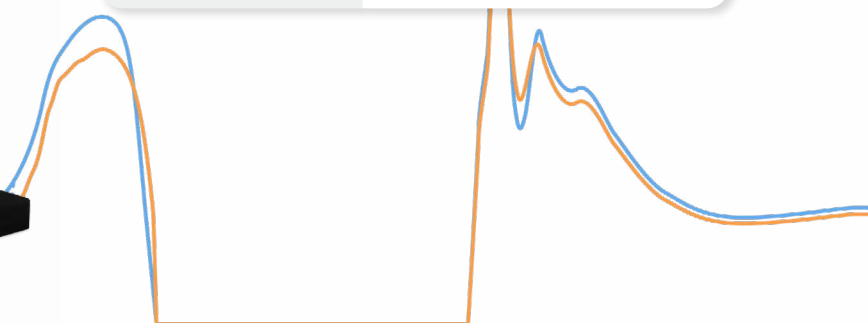
### Vertical Acceleration

$$\frac{\text{Force}}{\text{Mass} - \text{Gravity}} = \text{Acceleration}$$

Acceleration provides a measure of the rate at which vertical velocity changes. Vertical acceleration is dependent on gravity, the athlete's BM and the amount of vertical force they can produce.



| Metric                              | Insight                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------|
| Jump Height (IMP-MOM)               | Jump height achieved in the jump is calculated using the impulse-momentum formula.          |
| Concentric RFD (50ms, 100ms, 200ms) | Quantifies the speed of generating force at various time points from the start of movement. |
| Peak Takeoff Force Asymmetry        | Quantifies left vs. right differences in maximal force production.                          |



## 4. Practical Considerations

The SJ is often used alongside the CMJ in testing for two main reasons: to identify neuromuscular qualities not fully captured by the CMJ alone to compare with CMJ performance.

The SJ provides valuable insights into an athlete's concentric RFD. However, comparing RFD values across assessments (e.g., comparisons of CMJ maximum braking RFD, IMTP maximum isometric RFD and SJ maximum concentric RFD) are unlikely to be similar and should not be seen as equivalent measures of RFD.

Similar qualities like peak force and contraction times (along with RFD) are crucial for athletes like weightlifters and sprinters. For instance, while sprinting requires significant eccentric strength, the first few meters rely heavily on concentric force and RFD as athletes accelerate out of the blocks.

The SJ provides crucial insights into initial acceleration capabilities, making clear relationships to sprinting and running-based athletes. However, the SJ also provides quality insights into niche sports like Olympic weightlifting and powerlifting, where athletes must concentrically overcome a heavy load (barbell) from a static start, such as the clean or deadlift.

## Pitfalls & Considerations



### Test Consideration



A slight countermovement prior to initiating the jump is a common SJ compensation. For an excellent repetition, the force trace should only move upwards, as should the individual.

# Drop Jump (DJ)

## 1. Test Description

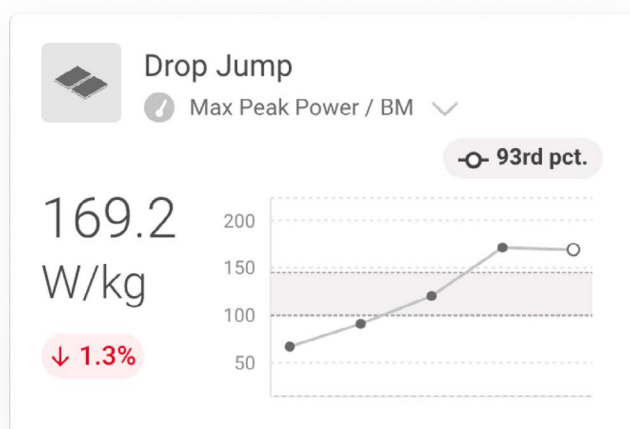
The DJ involves dropping from a fixed height and jumping as high and quickly as possible after making contact with the force plates. Due to the advanced technical and loading demands of this test, it may be difficult to execute well for individuals who have little plyometric exposure.

## 2. Test Insights

Reactive strength ability and plyometric capacity.

## 3. Common Test Metrics

- **Contact time:** Total time the individual's feet remain in contact with [ForceDecks](#) during the first landing (contact phase) of the test.
- **Jump height:** Maximum vertical displacement registered between the point of takeoff and the second landing.
- **Reactive strength index (RSI):** A ratio of the individuals' jump height divided by their contact time.



| Metric           | Insight                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------|
| Contact Time     | Total time the athlete's feet are in contact with ForceDecks during initial landing and takeoff phases. |
| Active Stiffness | Describes the relationship between peak forces and distance required to dissipate and create forces.    |
| RSI              | Jump height / contact time provides insights into reactive strength and jumping efficiency.             |

## 4. Practical Considerations

The DJ is a widely used test to assess an athlete's reactive strength or overall plyometric ability. It was popularized in the 1960s with [Soviet coach Yuri Verkoshansky](#) as part of his shock method training.

The DJ measures how well an athlete absorbs and reproduces force quickly. Key outcomes include contact time on the force plate, jump height and RSI, which is the most reported metric.

It is important to differentiate between a DJ and a depth jump, as depth jumps are inherently slower and not reflective of an athlete's reactive strength. To avoid accidentally performing a depth jump, ensure the individual understands the test outcomes, provide appropriate cues and ensure they have sufficient reactive strength ability, as demonstrated through hop testing trials or countermovement rebound jump (CMRJ).

The RSI, developed by [Warren Young \(1995\)](#) at the Australian Institute of Sport, is a cornerstone of plyometric training prescription and monitoring its effectiveness as part of the Strength Qualities Assessment Test (SQAT).

Creating a [DJ profile](#) helps determine the optimal DJ height for each athlete to ensure confident, actionable results. When testing longitudinally, it is crucial to ensure the drop height is the same across all tests for accurate comparison.

High RSI values in the DJ indicate an athlete's ability to produce quick, explosive movement that is crucial for repeated jumping in basketball or speed-specific sports like track sprinting.



### Test Consideration



Make sure to factor force plate height into box height calculations. ForceDecks are ~5.5cm (2.2in) in height from the floor and should be subtracted from the box height the athlete is dropping from for accuracy.

### Blind Spot: Depth Jump

Often mistaken with DJs, depth jumps primarily focus on jump height as the primary outcome, without considering jump strategy (e.g., contact time).



### Blind Spot: Drop Jump

DJs measure reactive strength based on the athlete's strategy to maximize their jump height while minimizing ground contact time (GCT).

# Hop Test

## 1. Test Description

In this repeated-effort plyometric test, drop height is controlled by the athlete's jump height. Unlike the fixed height box in the DJ, hop tests measure reactive strength based on the effective "drop height" calculated from the height of the first jump in a hop test.

Introduced by [Harper, Hobbs and Moore \(2011\)](#), the 10-5 hop calculation requires 10 hops, using the 5 repetitions with the highest RSI value for analysis. The hop test may be a more reliable introduction to reactive strength testing than a maximum-effort DJ.

It is important to note that while the 10-5 hop test protocol involves 10 hops, a higher or lower number of hops can also be used depending on the circumstances.

## 2. Test Insights

Repeated reactive strength abilities, plyometric capacity and potentially plyometric fatigability.

## 3. Common Test Metrics

- **Contact time:** Total time an individual's feet remain in contact with the ForceDecks through the contact phase of the DJ.
- **Jump height:** Maximum vertical displacement registered between each takeoff and landing point.
- **RSI:** A ratio of the athlete's jump height divided by their contact time.

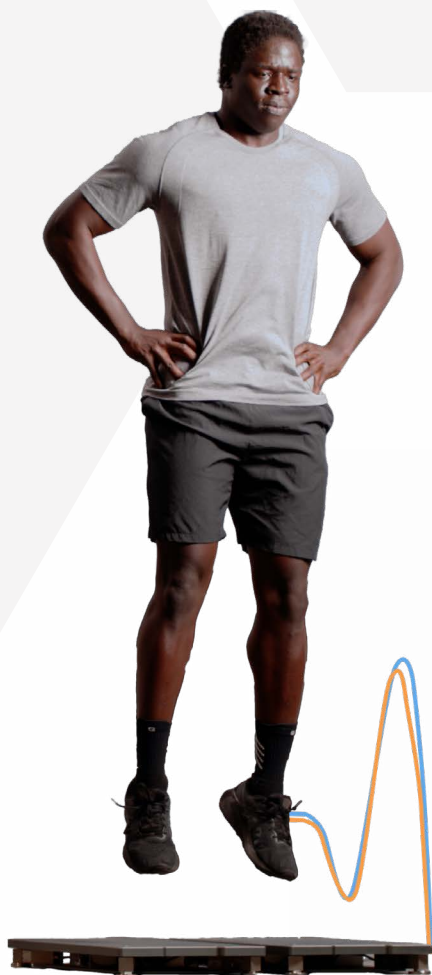
### Expert Tip:

During the hop test, the patient or athlete is only "falling" from heights they're capable of jumping, making the introduction to fast stretch-shortening cycle (SSC) activities more tolerable compared to a DJ, which tends to be more demanding.



**Ryan McLaughlin**  
Sport Scientist, VALD BDM

**VALD**



| Metric            | Insight                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------|
| Mean Contact Time | Average contact time during each hop. Provides a stable depiction of contact time.            |
| Mean RSI          | Average reactive strength performance over multiple hops. Reliable metric to track over time. |
| Mean Jump Height  | Average jump performance over multiple hops. Reliable indicator of repeated jump performance. |

## 4. Practical Considerations

The hop test, or 10-5 hop test, is a simpler alternative to the DJ for assessing reactive strength. Unlike the DJ, which requires one maximal effort from a set height, the hop test involves multiple hops, measuring the athlete's repeated efforts.

A higher RSI in the hop test reflects an efficient SSC, essential for quick vertical redirections. Field and court sport athletes, such as soccer or basketball players, with good RSI performance are often better equipped to land stiff, optimizing their utilization of elastic energy for SSC mechanics in sport.

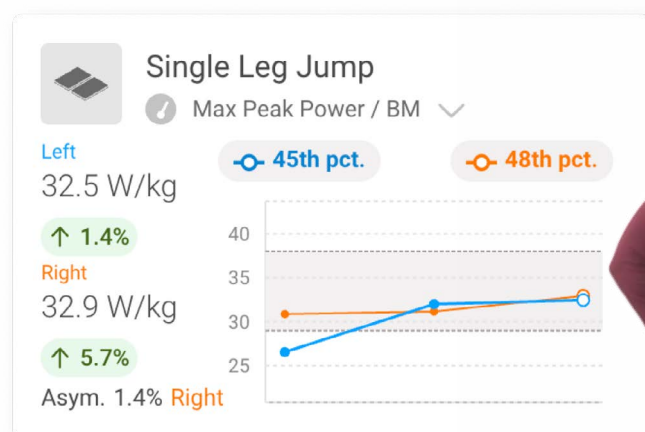
# Single Leg Jump (SLJ)

## 1. Test Description

The SLJ assesses unilateral power and jump performance. SLJs provide asymmetry measures that reflect an athlete's total lower limb capacity, unlike bilateral assessments, which reflect both capacity and compensatory strategies.

## 2. Test Insights

Maximal unilateral power, jump performance and single leg jumping strategy can be identified by assessing performance and asymmetries in the jump, specifically in the eccentric, concentric and landing phases.



### Expert Tip:

Athletes generate more impulse in SLJs with a contralateral leg swing, which significantly alters the force trace. For consistent comparisons, ensure the same jumping strategy is used throughout (leg swing vs. hip flexed).



**Gabriel Manoel**  
Physical Therapist, VALD CSM

VALD



## 3. Common Test Metrics

- **Jump height:** Maximum vertical displacement registered during the jump.
- **Phase-specific impulse (eccentric, concentric, landing):** Summed total force (force x time) within the eccentric, concentric or landing phase of the jump.
- **Eccentric peak velocity:** Maximum instantaneous speed of the countermovement descent.

## 4. Practical Considerations

A stable, quiet phase is critical for the SLJ, as it is with all tests. However, due to the balanced requirements of single leg stance, this is harder to achieve, leading many to compensate with one of two strategies:

- Lack of a sufficient quiet phase before executing the jump to reduce the likelihood of falling off balance.
- Using a hip, ankle or suspension strategy to maintain balance, altering the quiet phase force trace before the jump.

These strategies hinder movement detection, as individuals often start with slight knee flexion and perform shallow, slow descents during the SLJ. This can prevent proper eccentric phase detection, invalidating related metrics. Ensuring a smooth, flat force trace on one leg is essential before jumping.

In track and field, long jump performance relies on horizontal speed and vertical ground reaction forces from a single leg. High SLJ performance reflects efficient force application and hip-knee-ankle coordination, making it a valuable screening tool.

Similarly, in field and court sports, athletes often jump and land on one leg, requiring strong SLJ metrics to maintain stability and control. Consistent unilateral strength also supports safe landing mechanics.

| Metric                                           | Insight                                                                       |
|--------------------------------------------------|-------------------------------------------------------------------------------|
| Jump Height                                      | Overall indicator of jumping performance.                                     |
| Phase Specific Impulse (Ecc., Conc. and Landing) | Total amount of force produced within a specific phase of the jump.           |
| Peak Eccentric Velocity                          | Maximum speed of descent during movement. Good indicator of jumping strategy. |

# Countermovement Rebound Jump (CMRJ)

## 1. Test Description

Jumping and landing tests require individuals to perform rapid eccentric and concentric actions, assessing key physical components inherent to many athletic movements. The **CMRJ** combines two different jump types:

- a) an initial maximal effort CMJ, then
- b) an immediate rebound jump.

The CMRJ allows for the assessment of slow (CMJ) and fast (rebound jump) SSC mechanics in a single test. In other words, it tests the athlete's ability to perform both powerful and rapid movements – which are essential in sport.

The CMRJ can act as an alternative for the DJ test when executed correctly, with the key benefits being that the rebound portion is from a height that can be precisely calculated due to the initial jump, rather than from a standard box height.

## 2. Test Insights

Slow and fast SSC capabilities, maximal bilateral power generating capacity and reactive strength abilities.



### 3. Common Test Metrics

- **RSI:** A ratio of the individual's jump height divided by their contact time.
- **Contact time:** Total time the individual's feet are in contact with the force plates during the contact phase of the DJ.
- **Rebound jump height:** Maximum vertical displacement registered after the second point of takeoff and the second landing.

### 4. Practical Considerations

When performing the CMRJ, all individuals should be familiarized beforehand, ideally in a separate test session or by regularly including the movement in the sport-specific warm-ups.

It is also important to standardize the test protocol (hands on hips is recommended) and cueing instructions. A simple approach is to cue athletes to jump "fast and high."

As with the DJ, it is suggested that the GCT should be < 250ms in the rebound portion of the jump. Longer contact times invalidate the assessment of fast SSC and reactive strength and suggest either the test has not been cued correctly or the individuals lack the required physical capacity to effectively perform the movement.



#### Blind Spot: Environmental Standardization

Consistent testing environments are essential for reliable data. If the force plates need to be moved, they must be re-zeroed before testing to avoid errors, ensuring stable conditions for every assessment.

# Balance Tests

## Key Consideration when Administering Balance Tests

Where traditional balance tests relied simply on time or subjective measures of stability, modern force plates can accurately measure center of pressure (CoP) and provide precise and objective assessment of balance.

By tracking subtle CoP shifts, force plates offer detailed insights into stability, including directional control and velocity. This technology uncovers balance nuances and asymmetries

that may be missed with subjective observation, allowing for accurate monitoring and tailored interventions to support rehabilitation, injury prevention and performance optimization.

Selecting the correct level of challenge is crucial. For those who struggle with single leg balance, a bilateral test like the quiet stand may be more appropriate before progressing to single leg assessments.

## Practical Considerations for All Balance Assessments

Balance assessments are often performed with eyes open and closed to evaluate balance strategies and visual system reliance. Modern force plates can also measure CoP through unstable surfaces, such as foam pads.

This capability allows practitioners to assess balance strategies and sensory responses in varying conditions, offering a more comprehensive understanding of balance performance.

As with all balance assessments, placing force plates in a corner or with a chair in front of the patient or athlete can be helpful risk mitigation strategies.

The walls allow for safe feedback if postural sway exceeds their base of support, and the chair can be used for external support if needed.

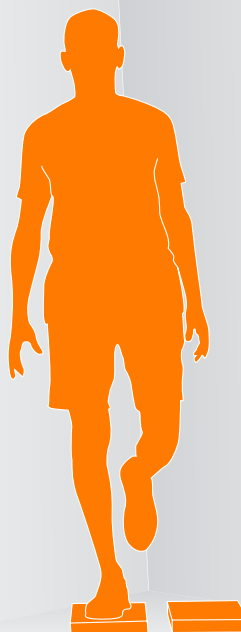
### Expert Tip:

Before the weighing period, placing a foam roller upright on the opposite plate allows you to track the number of times clients reach for external support without losing balance. Placing a foam pad underneath can add extra challenge while still providing accurate results!



**Jamie Youngson**  
S&C Coach, VALD Client Success Manager

**VALD**



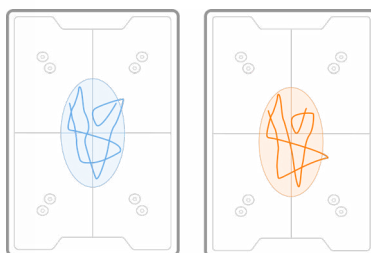
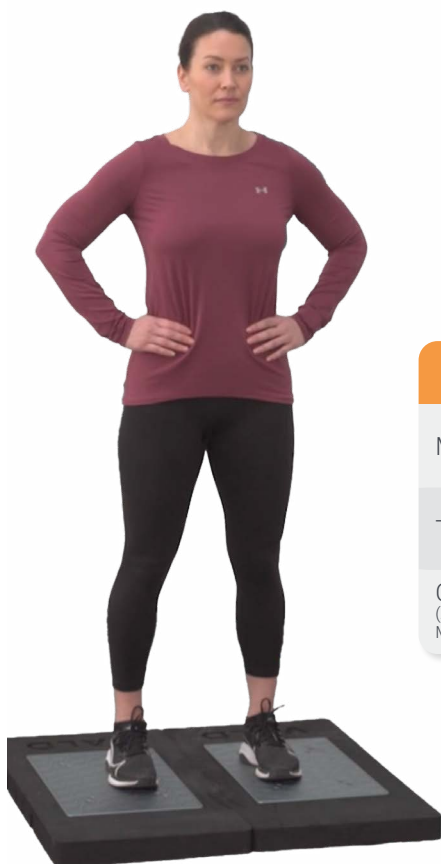
# Quiet Stand

## 1. Test Description

A bilateral balance test to assess the participant's balance strategy in a normal standing posture.

## 2. Test Insights

Allows the practitioner to identify the strategy the participant uses to maintain this position and if there is a lot of instability or if the participant can perform the test with relative ease.



| Metric                                          | Insight                                                                                          |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Mean Velocity                                   | Measured in millimeters per second, represents average speed of CoP sway during the test.        |
| Total Excursion                                 | Total distance CoP travels during the test. Helpful indicator of stationary bilateral stability. |
| CoP Range<br>(Anteroposterior and Mediolateral) | Range of distance covered in cardinal planes during the test, shows directional bias.            |

## 3. Common Test Metrics

- **Anterior-posterior and medial-lateral CoP ranges (mm):** The distance traveled by the CoP along the anterior-posterior (front-to-back) axis during a specific test or movement. This measurement is also taken for the medial-lateral (side-to-side) axis.
- **Mean velocity (mm/s):** Average speed the CoP moves over the course of a balance assessment.
- **Total excursion (mm):** Total distance traveled by the CoP over the course of a balance assessment.

# Single Leg Stand

## 1. Test Description

A unilateral balance test assesses the participant's balance strategy in a single leg standing posture.

## 2. Test Insights

Allows the practitioner to identify the strategy the participant uses to maintain this position and if there is a lot of instability or if the participant can perform the test with relative ease.



| Metric                                          | Insight                                                                                          |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Mean Velocity                                   | Measured in millimeters per second, represents average speed of CoP sway during the test.        |
| Total Excursion                                 | Total distance CoP travels during the test. Helpful indicator of stationary bilateral stability. |
| CoP Range<br>(Anteroposterior and Mediolateral) | Range of distance covered in cardinal planes during the test, shows directional bias.            |

## 3. Common Test Metrics

- **Mean velocity (mm/s):** The distance traveled by the CoP along the anterior-posterior (front-to-back) axis during a specific test or movement. This measurement is also taken for the medial-lateral (side-to-side) axis.
- **Total excursion (mm):** Total distance traveled by the CoP over the course of a balance assessment.
- **Excursion asymmetry between limbs (%):** Percentage increase of the lower performing side, compared to the steadier side, when measuring the total excursion of the CoP.

# Single Leg **Range of Stability**

## 1. Test Description

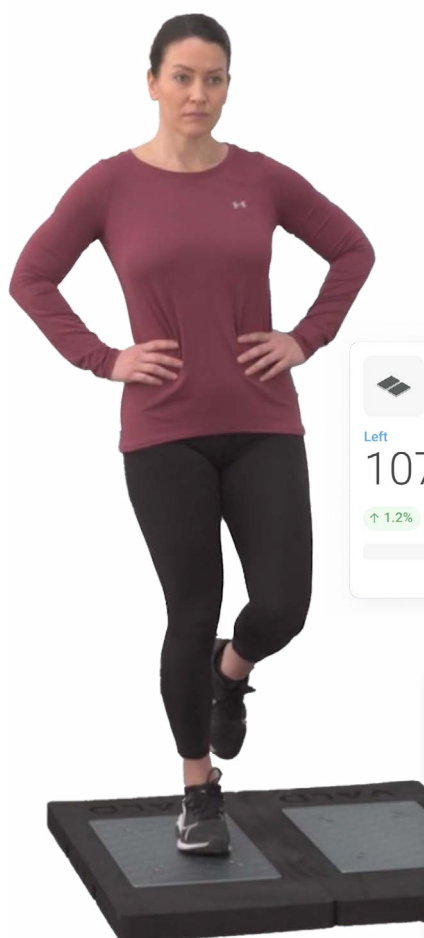
Assesses balance through a dynamic ROM, where the participant maintains a single leg stand and then leans as far as possible in each direction (forward, backward, left, right).

## 2. Test Insights

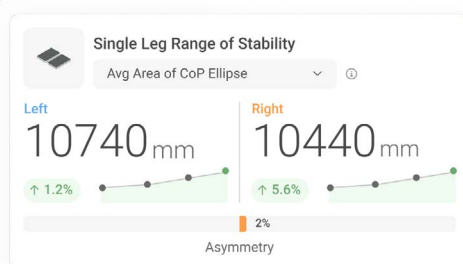
The strategy is employed to maintain balance while moving through this dynamic movement.

## 3. Common Test Metrics

- **Anterior-posterior and medial-lateral CoP ranges (mm):** The distance traveled by the CoP along the anterior-posterior (front-to-back) axis during a specific test or movement. This measurement is also taken for the medial-lateral (side-to-side) axis.
- **Total excursion (mm):** Total distance traveled by the CoP over the course of a balance assessment.
- **Area of CoP ellipse (mm<sup>2</sup>):** The surface area, measured in an ellipse, on [ForceDecks](#) that encompasses 95% of the data points measured during a balance assessment.



| Metric                                       | Insight                                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Area of CoP Ellipse (mm <sup>2</sup> )       | Measured in millimeters squared, represents average spread of CoP sway during the test.             |
| Total Excursion                              | Total distance of CoP travels during the test. Helpful indicator of stationary bilateral stability. |
| CoP Range (Anteroposterior and Mediolateral) | Range of distance covered in cardinal planes during the test. Shows directional bias.               |



### **Blind Spot: Force Plate Zeroing**

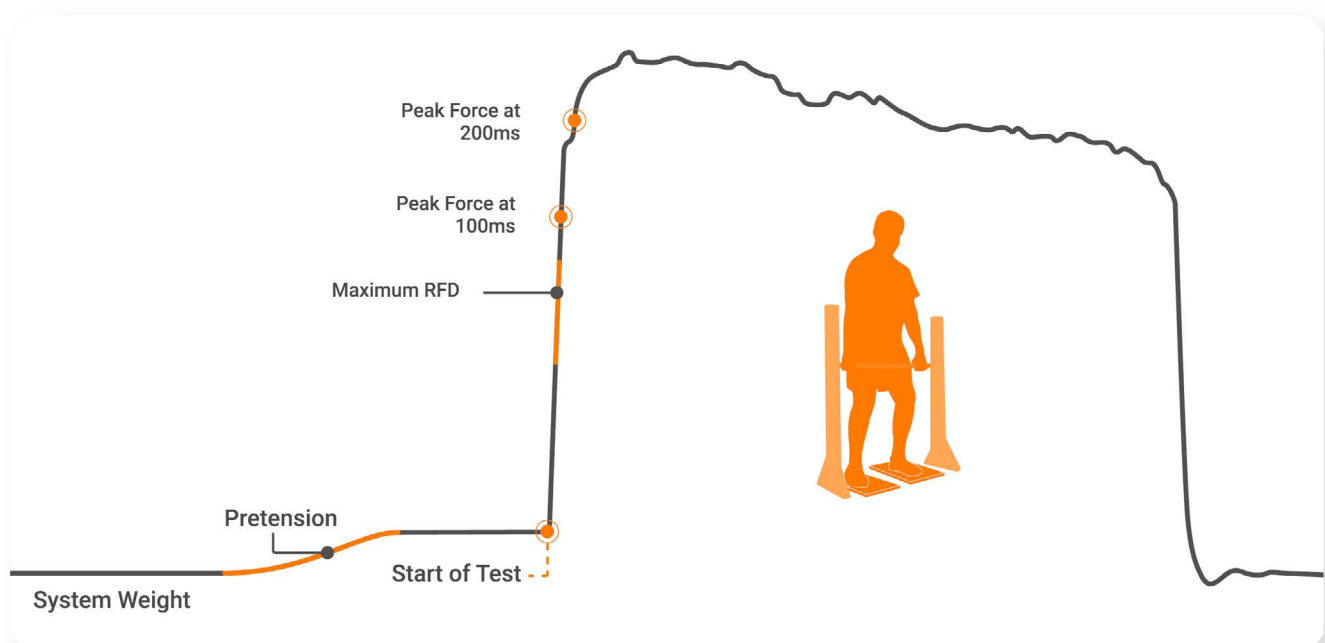
Proper zeroing is critical. Any mishandling here compromises all subsequent data. Ensure nothing touches the plates during zeroing, and be aware of environmental factors like vibrations. In a busy gym, even a nearby barbell drop could disrupt this process. Remember: "If in doubt, zero it out."

# Advanced Metrics, Indexes and Profiles

## Rate of Force Development (RFD)

**RFD** refers to how quickly force is generated, making it a key metric in assessing an athlete's explosive strength capabilities and a critical attribute in all explosive movements.

RFD is calculated as the change in force over a specified time interval relative to the onset of force production, e.g., force developed over the first 100ms of the maximal effort contraction. RFD can also offer insights into an athlete's neuromuscular readiness, as it captures the athlete's ability to generate force rapidly, which may be a more sensitive marker of response to load than peak force.



Practitioners should be aware that RFD can be challenging to measure reliably in some individuals and testing environments, resulting in high variability between trials and the influence of technique or fatigue.

Therefore, metrics like **peak force** at a **specific time point** are often favored for their greater reliability.

For example, ForceDecks provide “force at time point” metrics to represent RFD at intervals of:

- 50ms;
- 100ms;

- 150ms; and
- 200ms.

Typically, 0-100ms and 200ms metrics are selected as windows corresponding to the RFD-related capacities of the neural and muscular systems.

Unlike RFD, which is highly sensitive to changes within small time windows, peak force at a specific time can give a more consistent snapshot of the individual's maximal force output at a key moment, minimizing measurement noise and enhancing the robustness of data across sessions.

# Dynamic Strength Index (DSI)

## 1. Test Description

The DSI is a ratio that compares similar metrics from two separate assessments. The DSI compares an athlete’s maximal ballistic strength to their maximal isometric strength to provide simple testing metrics to help guide training decisions.

The DSI compares an athlete’s maximal isometric strength (via the IMTP test) to their ballistic strength (via the CMJ test), providing valuable insights into the balance between these two capacities.

## 2. Applications

Practitioners use the DSI to quantify force output characteristics and aid decision-making if the individual would benefit more from focusing on maximal strength development or ballistic strength development.

This ratio highlights an athlete’s capacity to translate maximal strength into explosive movement. Unlike isolated measures, DSI delivers a more nuanced profile of an athlete’s force capabilities, guiding targeted programming to optimize both performance and injury resilience.

## 3. Test Insights

Based on the results of the DSI, practitioners can determine an athlete’s ability to produce force and make decisions around the direction of training for their athletes and patients.

## 4. Assessment Strategies Metrics

- Peak force applied in a ballistic test (e.g., CMJ).
- Peak force applied during a maximal isometric test (e.g., IMTP).

## 5. How to Calculate the DSI

Ballistic Strength ÷ Isometric Strength



## 6. Interpreting the Data

| Outcome   | Recommendation                           |
|-----------|------------------------------------------|
| < 0.6     | Ballistic strength training recommended. |
| 0.6 – 0.8 | Concurrent training recommended.         |
| > 0.8     | Maximal strength training recommended.   |

## Practical Considerations for DSI

When comparing maximal effort outputs between two assessments, an individual may have more experience with one assessment than another. This familiarity may skew the data to show a larger discrepancy between outputs than truly exists. Ensure the athlete is highly familiarized with both assessments before calculating the DSI.



### DSI

#### Dynamic Strength Index

A ratio between an athlete's maximum isometric strength and their ballistic strength ability.

**Limitations**  
Nonspecific assessment of general qualities.  
Experience with isometric tests can change outputs.

Tests: **CMJ** **IMTP**

Calculation: **Ballistic Strength (N)** ÷ **Isometric Strength (N)**

Example: **2030N** ÷ **2209N** = **0.92**

Results:

|                                        |                                         |                                      |
|----------------------------------------|-----------------------------------------|--------------------------------------|
| <b>&lt;0.6</b><br>Emphasize Ballistics | <b>0.6 - 0.8</b><br>Concurrent Training | <b>&gt;0.8</b><br>Emphasize Strength |
|----------------------------------------|-----------------------------------------|--------------------------------------|

## Eccentric Utilization Ratio (EUR)

### 1. Test Description

EUR provides a ratio between CMJ and SJ performance, specifically, the height achieved in both.

### 2. Test Insights

This test provides insights into the ability of the athlete to utilize their SSC while jumping.

As the CMJ involves a countermovement and some eccentric bias, we typically expect the athlete to achieve a higher jump height than in the SJ test, which is a concentric-only jump.

If an athlete does not jump higher in the CMJ than the SJ, there may be an issue with how they are leveraging their SSC.

### 3. Calculation

CMJ Height (m) / SJ Height (m)

### 4. Interpreting the Data

A ratio of **1.1:1** indicates an athlete who is adequately able to utilize their SSC. An example of a 1.1:1 ratio would be a CMJ height of 40cm (15.8in) and an SJ height of 36cm (14.2in).

## Practical Considerations for EUR

Familiarization with both testing protocols is crucial to get an accurate insight into someone's EUR. Both the CMJ and SJ need to be performed to an excellent level.

It is important to consider that while the goal of achieving a high EUR may be an applicable goal across the majority of athletic populations, some sports may benefit from achieving higher numbers in the SJ test than in the CMJ, such as powerlifters.



### EUR

**Eccentric Utilization Ratio**

A ratio between an athlete's CMJ height and SJ height. It provides insight into how well an athlete utilizes their stretch-shortening cycle.

**Limitations**  
The athlete needs to be familiarized with both tests.

Tests: SJ CMJ

Calculation: CMJ Height (m) ÷ SJ Height (m)

Example: 0.41m ÷ 0.40m = 1.03

Results: <1.0 1.0 - 1.1 >1.1

## Drop Jump Height Profile

### 1. Test Description

A DJ profile is a progressive DJ assessment of increasing height incrementally while tracking key metrics, such as GCT or RSI.

### 2. Test Insights

Facilitates an optimal height of a box for training the DJ to best elicit adaptation for the individual.

### 3. Common Test Metrics

GCT, RSI

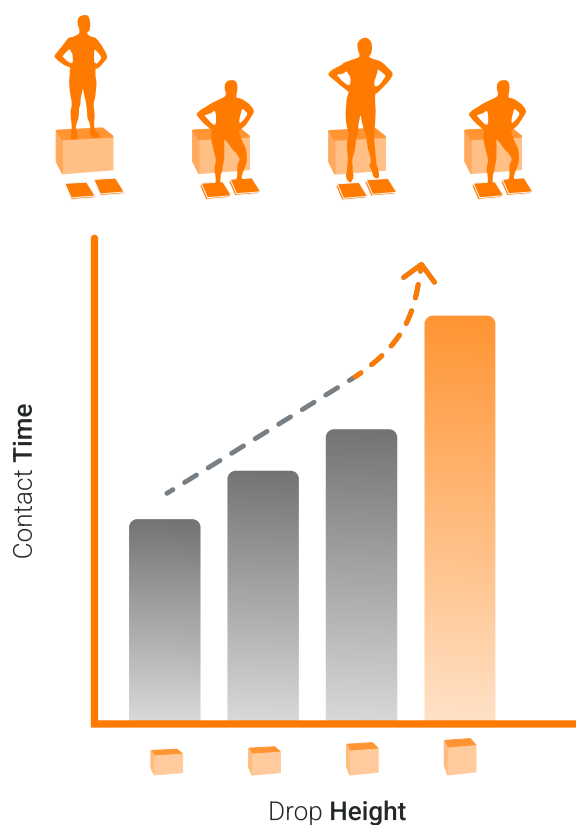
### 4. Interpreting the Data

If a typical GCT for an individual's DJ is around 190ms ±5%, there will likely be a threshold where GCT increases as box height increases. If, for example, there is a 10% increase in GCT (>209ms) on a given height, this height may be used as a cut-off or "ceiling" for DJ training and future assessment.

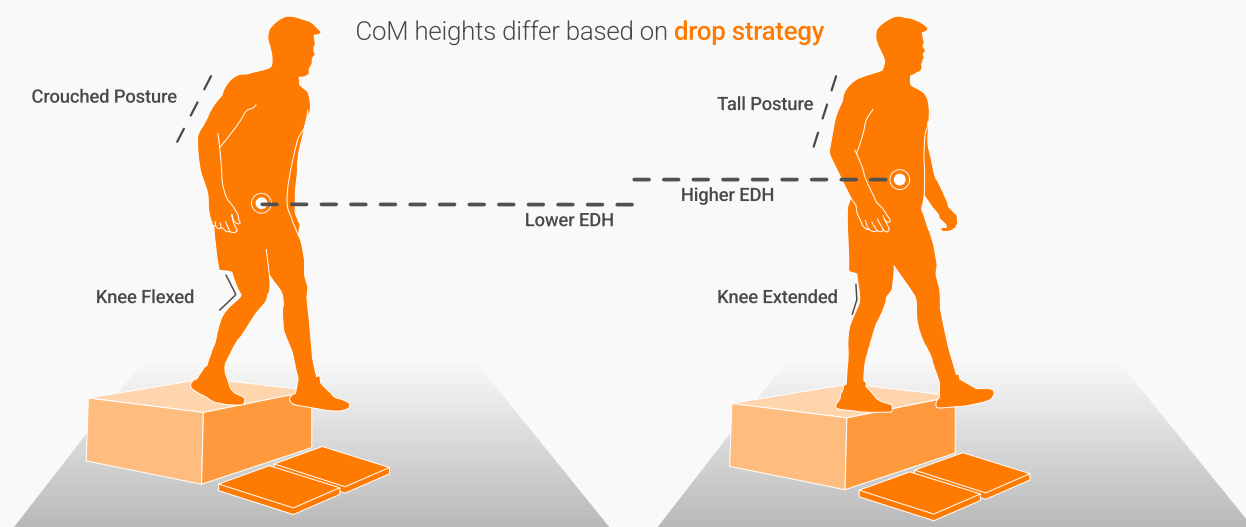
When progressively increasing DJ height during the DJ profile, it is important to check effective drop height. When dropping from a box, the individual may lower their center of mass (CoM) before stepping off the box, decreasing their effective drop height. Make sure individuals consistently drop off the box with the same strategy each time to ensure adequate comparisons.

## Practical Considerations for DJ Height Profile

This assessment is a highly demanding activity, and practitioners should ensure that individuals are adequately prepared and have significant previous plyometric exposure. Individuals should employ the same strategy, with the appropriate cueing, for each DJ as the height increases to get a true measure of reactive strength capability.



### Effective Drop Height (EDH)



# Force Velocity Profile (FVP)

## 1. Test Description

An FVP with force plates (also known as a vertical FVP) investigates the relationship between an athlete's speed of vertical movement with incremental loading changes.

Typically, this is done through vertical jump testing with increasing loads (e.g., 10% BW, 15% BW, 20% BW).

## 2. Test Insights

Simply, if you improve an athlete's force and velocity abilities during a jump, their outputs (e.g., jump height) should improve. Understanding which component to focus on is where FVP becomes valuable. FVPs identify the strengths and weaknesses that individuals have between slow, high-load jumps and fast, low-load jumps.

These insights help to inform if an athlete excels more with heavy loads (termed "force dominant") or if they excel more with fast speeds (termed "velocity dominant"), depending on the results of the FVP.

## 3. Common Test Metrics

FVPs have typically been conducted with jump height as the primary metric. However, with the integration of force plates, you also have access to metrics such as peak force (N) or relative force (N/kg), velocity at takeoff (m/s) and peak power (W or W/kg).

For those wanting to dive deeper into metric selection for FVP, investigating the relationship between these metrics may be helpful when conducting an FVP with athletes.

## 4. Interpreting the Data

By plotting force and velocity across a range of loaded and unloaded jumps, practitioners can determine if an athlete's profile shows higher performance with heavier loads and slower speeds (force-dominant) or with lighter loads and faster speeds (velocity-dominant).

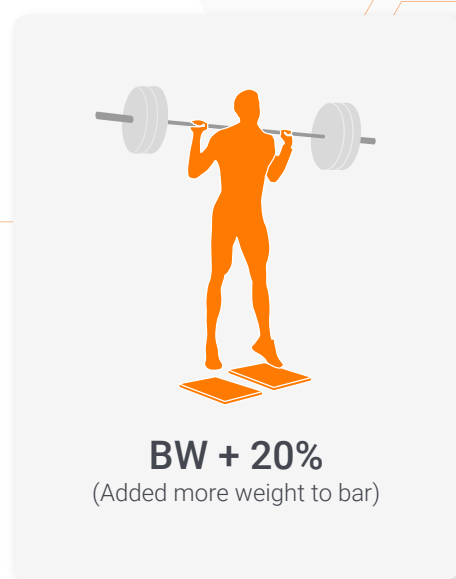
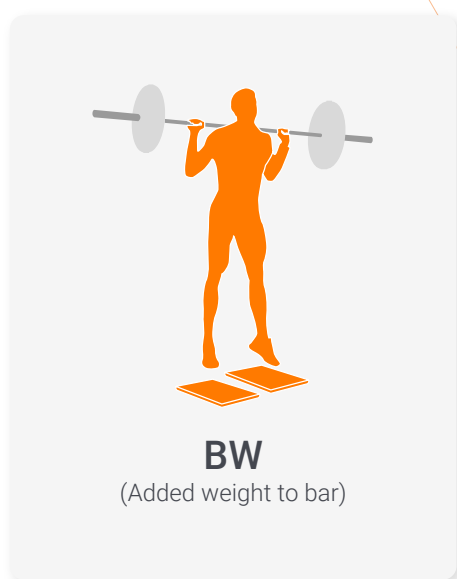
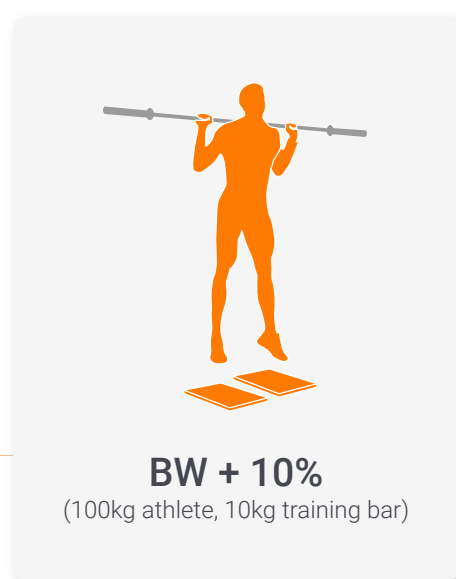
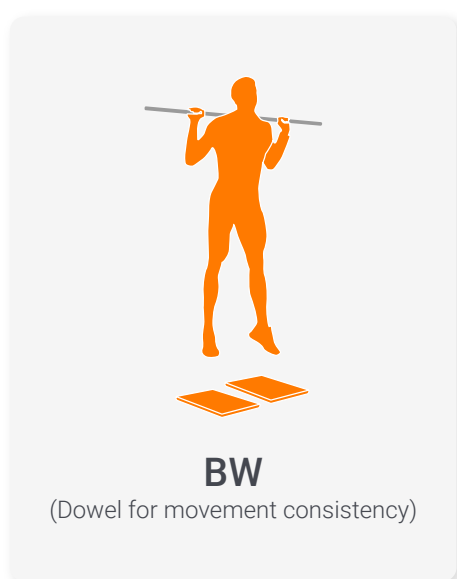
For example, force-dominant individuals may benefit from training periods focusing on light loads and higher velocity training strategies to "balance" their FVP. On the other hand, velocity-dominant athletes might benefit more from slow, heavy lifts to improve their maximal force expression.

Practical applications of these insights include tailoring training interventions to optimize the athlete's force-velocity balance, which may lead to improved power output and help make a more well-rounded athlete.

## Practical Considerations Considerations for FVP

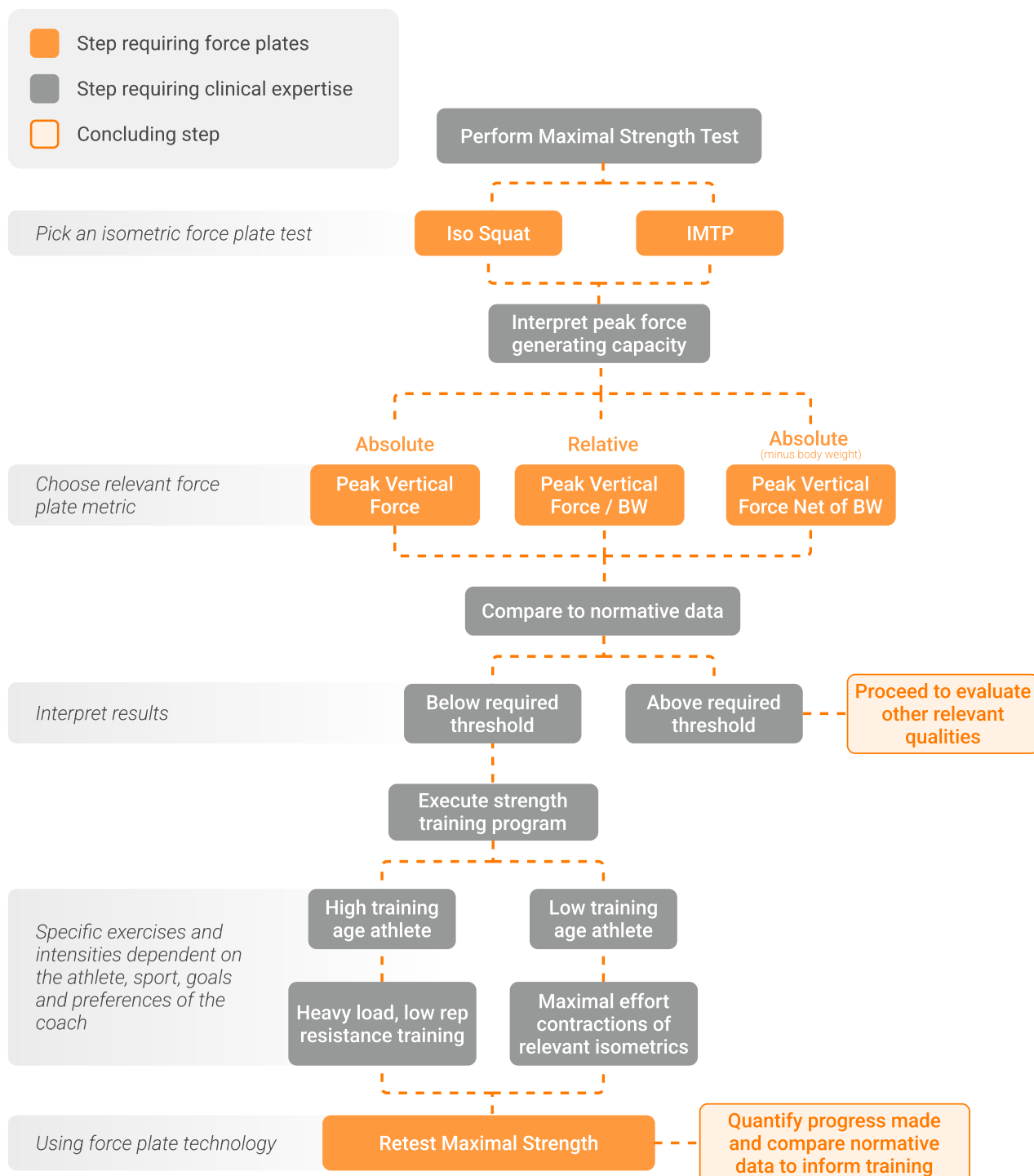
FVPs are highly specific to the movement tested and may not fully reflect the demands of a given sport. For example, controlled vertical jumps are useful but inherently different from jumping tasks seen in sport.

These differences may explain why some research shows world-class athletes have seemingly “sub-optimal” FVPs, suggesting extensive practice in a sport may create unique, sport-specific force-velocity relationships that differ from a typical FVP assessment.



# Concepts in Force Plate Interventions

## Developing Strength



A practitioner can determine whether an athlete's force-generating capacity is sufficient by comparing common isometric assessment results to normative data or their own baseline.

While region-specific force plate strength assessments exist, such as ASH testing and RSISA, more common measures of total body strength include the IMTP and Iso Squat.

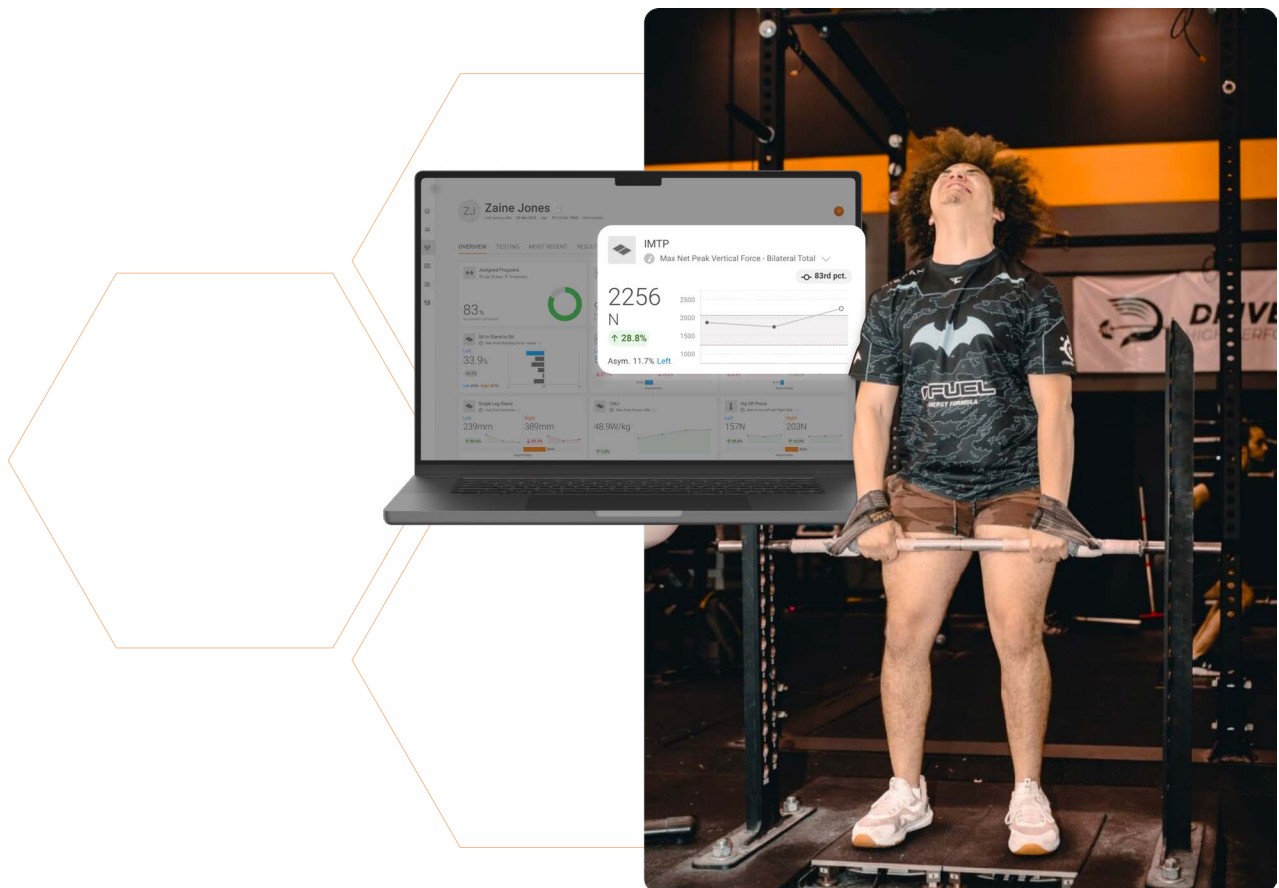
If values are lower than expected, programming maximal strength training can reduce the strength discrepancy.

Maximal strength development involves high-load resistance training to stimulate neuromuscular adaptation. Typically, 1-5

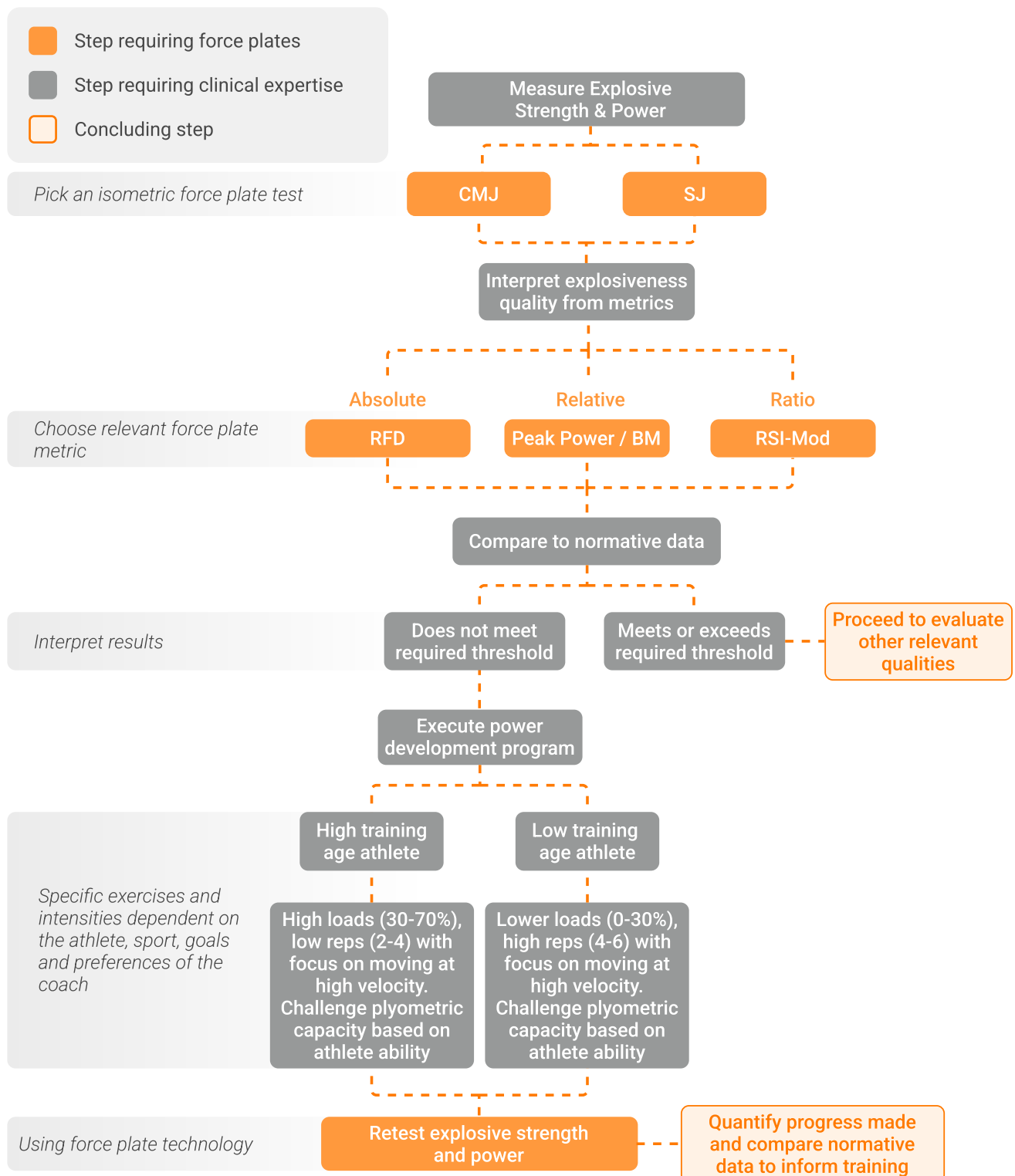
repetitions with heavy loads near an athlete's maximum capacity are required, though moderate loads can also be effective.

The training frequency and volume should align with the athlete's capacity and competitive schedule, with more volume in the preseason and minimal effective doses during the season.

Advanced techniques like supramaximal eccentric training, isometrics, variable resistance (typically involving chains or bands) and partial repetitions (limited ROM) can further enhance strength and force production.



# Developing Explosiveness



Maximal strength development can significantly enhance RFD, contraction times and power output. Some athletes may be “strong enough” but still lack the necessary RFD properties for their sport. Objectively weaker athletes will often show deficits in both peak force and RFD. They should, therefore, prioritize maximal strength, but concurrent strength and power-based training can also be effective.

If assessments reveal deficits in explosive strength – such as low CMJ height or slow 10m acceleration – incorporating ballistic training with varying loads can boost power output.

This may include Olympic lifts and their derivatives and moderate to light jump-based movements. For athletes who meet normative strength levels, greater emphasis can shift toward power development through exercises like Olympic lifting variations and plyometrics.

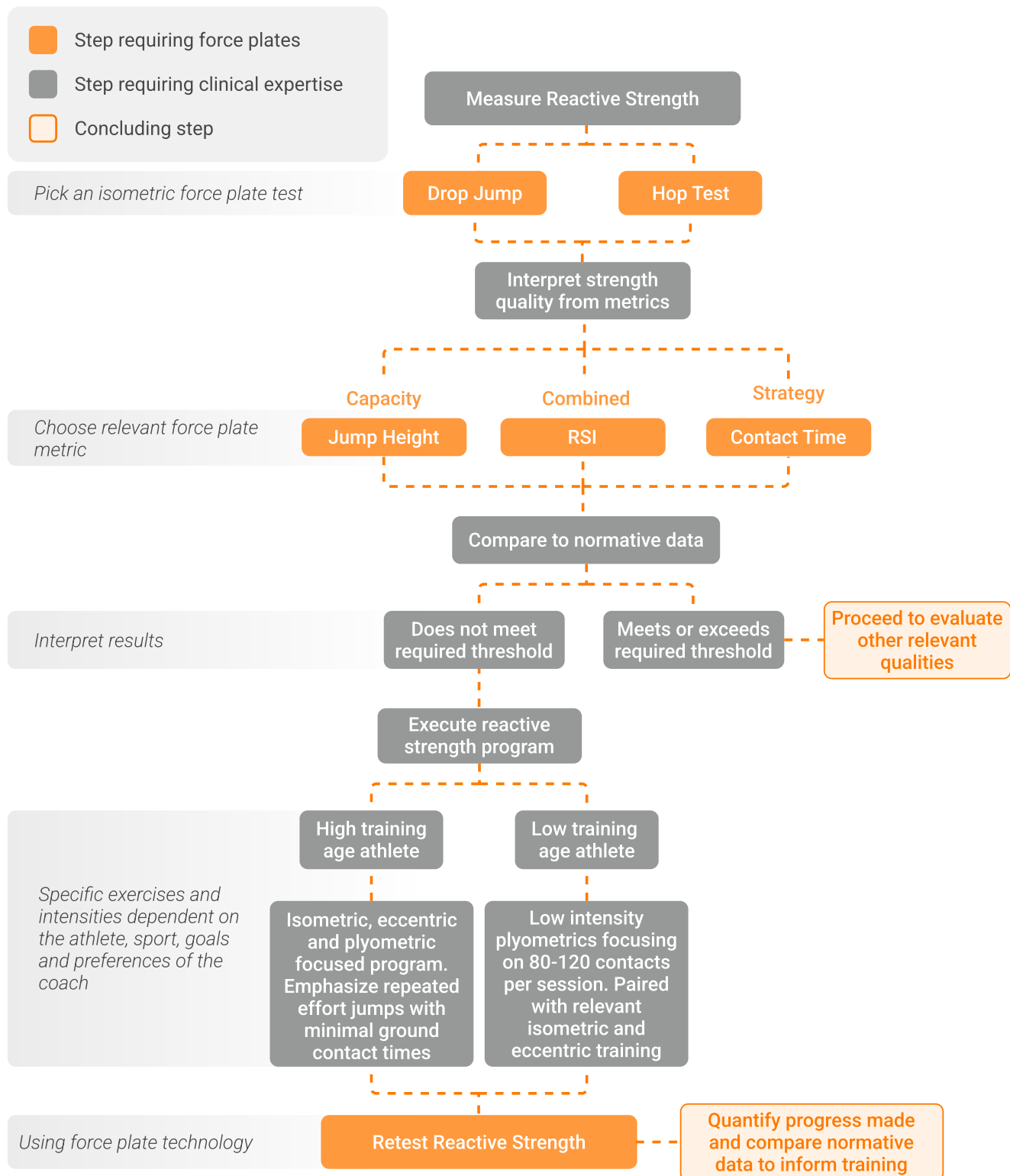
Muscular power can be trained with loads ranging from light (<30% 1RM) to heavy (>70% 1RM). However, key variables such as movement velocity and maximal effort or intent are key drivers of power training.

Utilizing “optimal power loads” are athlete-specific and are often attained through [force-velocity-power profiling](#). Periodization is key: early phases may focus on maximal strength with concurrent low-load and high-velocity exercises, while later phases may shift toward power development, reflecting competitive demands.

Depending on the loading scheme, power training can take place 3-4 times per week, with ballistic exercises typically involving 2-6 sets of 1-6 repetitions. Plyometrics and jump-based movements may include 3-6 sets of 6-10 repetitions, 2-3 times per week.



# Developing Reactive Strength



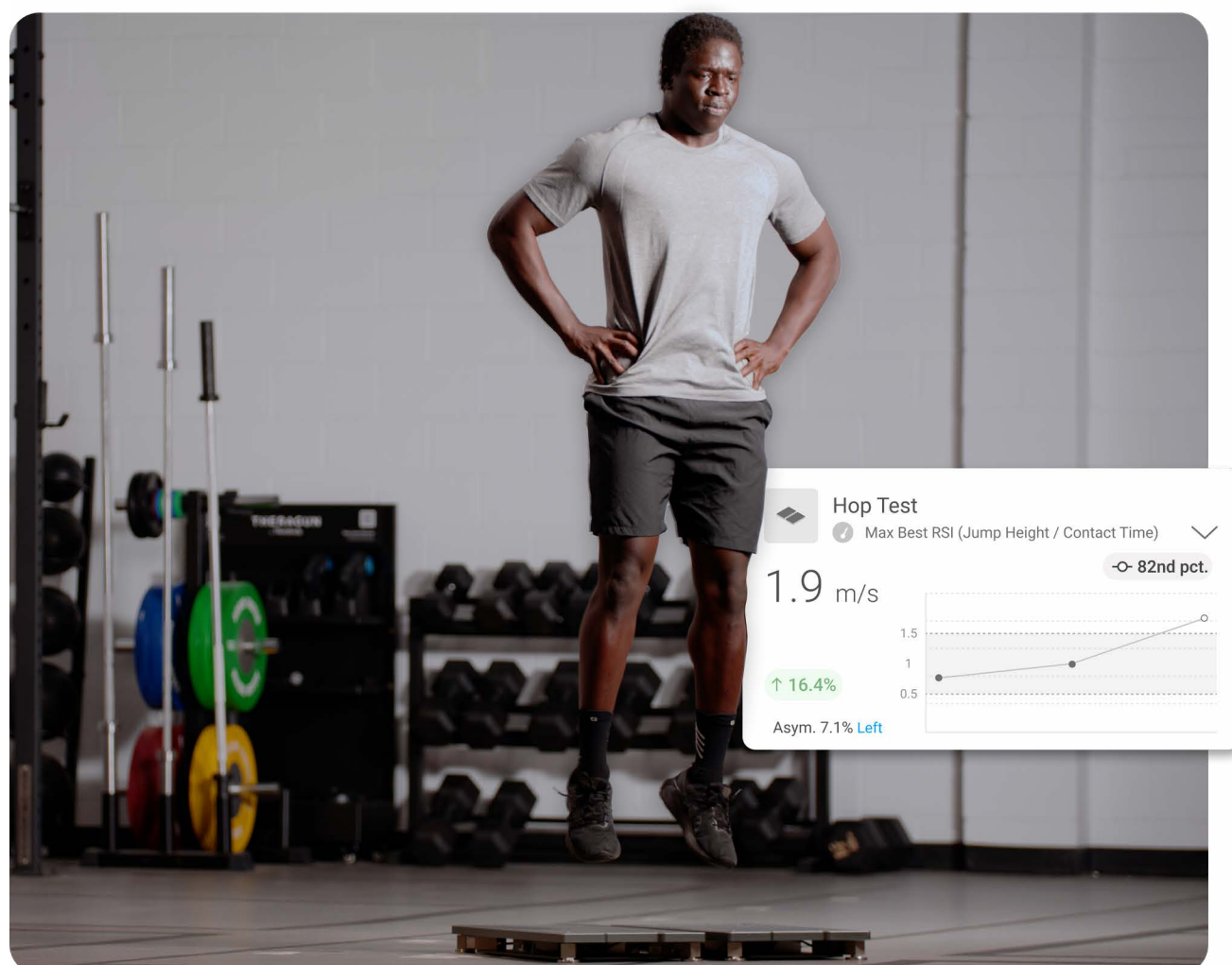
Reactive strength and lower limb stiffness can be developed through various training methods. Maximal strength and power training, for example, improve reactive strength indirectly via the nervous system and soft tissue adaptations, like increased nerve conduction velocity and improved tendon stiffness, respectively.

Individuals with extensive training experience will need more targeted interventions to enhance reactive strength. If a significant deficit is detected (e.g., [contact time longer than 225ms in a DJ](#)), plyometric exercises are highly effective. Plyometrics enhance reactive strength by utilizing elastic energy stored in tendons, which play a crucial role in force transfer.

To improve reactive strength, combine isometric strength training with plyometric training, as isometrics can increase tendon stiffness. Plyometric exercises like DJs, hurdle hops and bounding require a baseline of general and eccentric strength to handle the high eccentric demands. Eccentric strength training is beneficial in preparing for these exercises.

Jumps and plyometrics are suitable for athletes at any stage, provided the exercises are scaled appropriately. Athletes can perform lower-intensity plyometrics 2-3 times per week during general preparation periods, with 1-2 sessions per week during competition.

Typical volumes range from 80-140 contacts in preparation, while competitive phases might involve a microdose of 10-20 contacts throughout the microcycle. Due to the high impact forces of plyometrics, careful balance with other training loads is essential.

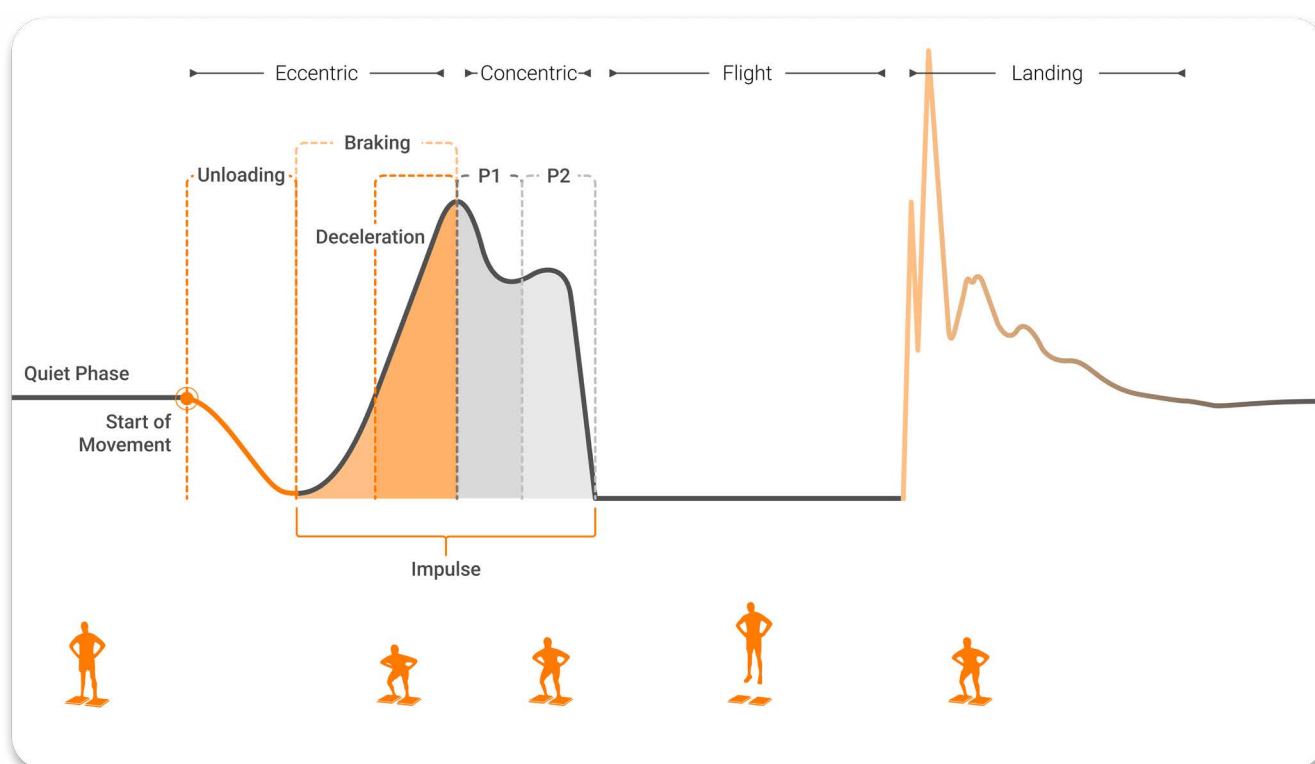


# Force Trace Analysis

The force trace shows the practitioner what the performance of a jump or isometric test looks like. Being able to interpret this allows the practitioner to quickly decide if the athlete performed the test to an adequate level and whether it should be included in the data analysis or if it should be discarded. If that is the case, the practitioner and athlete should repeat the test with additional coaching cues.

## Countermovement Jump

Below is a force trace from a CMJ that is sufficient to include in data analysis.



**Quiet Phase:** This phase occurs before the downward movement begins prior to every rep. A short, steady quiet phase prior to the start of movement – affects the accuracy of a number of valuable metrics.

A common issue observed is individuals shuffling their feet just before the jump or upper body movement, both of which create noise affecting the vertical force trace, causing shifts between left and right force traces.

Practitioners should keep an eye on the left and right force traces while they are giving instructions; both of these should be parallel and smooth.

**Unloading Phase:** This begins after the start of movement, where the force trace drops as the athlete descends. Ideally, the trace should drop steeply, whereas a shallow trace suggests improper technique, such as hinging at the hips rather than bending at the knees or a very slow descent this

may be due to poor tolerance for the upcoming deceleration – such as if experiencing knee pain or during rehab – or potentially because they have not understood the instructions on jump technique.

**Braking Phase:** The point at which the force trace begins to increase marks the start of the braking phase. This phase continues until zero velocity (the lowest point in the movement). Braking performance indicates how quickly the athlete applies force, with steeper traces being ideal.

**Eccentric Deceleration Phase:** Beginning at eccentric peak velocity as the individual continues to descend but decelerating. This part of the braking phase begins when the force trace crosses BW and continues until zero velocity. It measures how effectively the athlete vertically decelerates, an important factor for tasks like running and jumping. In monitoring environments, increased durations can indicate neuromuscular fatigue while shorter durations suggest a positive response to training or loading.

**Zero Velocity:** This is the transition from downward to upward movement, where the eccentric phase ends and the concentric phase begins. Higher forces at zero velocity are desirable as they indicate strong deceleration capacity and contribute to jump height performance.

**Concentric Phase:** Begins after zero velocity and continues until the body leaves the force plates (defined as when force drops below 20N).

**Concentric Phase 1 (P1):** This measures force during the first 50% time-wise of the concentric phase. It reflects the athlete's ability to generate force from the deepest part of the countermovement and the hip-knee dominant transition out of the triple-flexed position.

**Concentric Phase 2 (P2):** The second half of the concentric phase focuses on propulsive ability and triple extension, including plantarflexion or displacement at takeoff. Individuals returning from anterior cruciate ligament (ACL) injury often have lingering P2 deficits and elevated asymmetries.

**Flight Time:** The time between takeoff/toe-off and landing detection (force above 20N). Flight time is one of the methods used to estimate jump height and part of the FT:CT ratio. While the IMP-MOM method is often preferred for a more accurate CoM displacement, there are conditions, such as where accurate BW measurement has not been performed, that would promote the use of the flight time calculation instead.

**Landing Phase:** This measures how well the athlete decelerates upon landing, factoring in jump height, BW and landing proficiency.

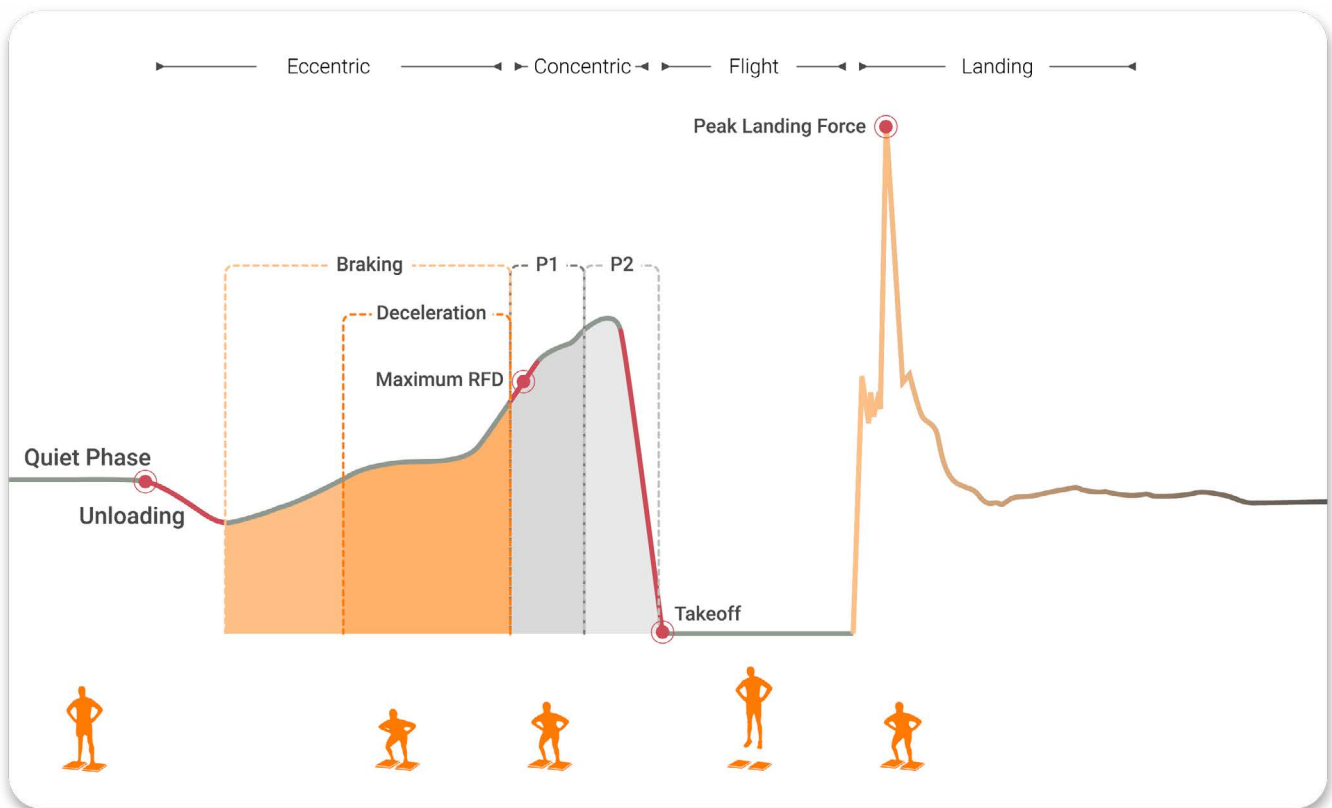
#### Expert Tip:

"Jump high" versus "jump high and fast" cues result in different outputs for a range of metrics. The "AND as fast" cue increases eccentric peak velocity. An adequate eccentric peak velocity is critical to obtaining eccentric force, RFD and power metric values representing true capacity. 1.2ms eccentric peak velocity is a general minimal target, but expect lower values and higher variability in early rehabilitation.



**Daniel Cohen, PhD**  
Co-Founder of ForceDecks

Due to the accessibility and ease of implementation of the CMJ, many athletes and patients will perform the movement with various strategies and means of maximizing their performance. Below is a force trace of a suboptimal CMJ. The outlined areas illustrate how to assess and understand how the trace can uncover deficits associated with vertical jump performance.



**Unloading:** The unweighting phase of the CMJ is crucial, as it drives momentum and overall jump performance. Athletes who struggle to unload quickly often compensate with greater concentric force.

Practitioners should examine force registered at the start of the braking phase – ideally, minimal force or a lower value here signals an effective unloading strategy.

**Eccentric Performance:** Even with a small amount of unloading, the athlete takes a significant amount of time to brake and decelerate their momentum in the downward or eccentric phase. This deceleration strategy is not quick enough to elicit an SSC response, negatively contributing to the athlete's jumping performance.

**Maximum RFD:** The individual's maximum RFD occurs in the concentric phase of the movement. While not inherently a negative performance indicator, basic muscle physiology dictates that maximum concentric contractions have less capacity than maximum eccentric contractions.

If peak RFD occurs in the concentric phase, this is likely a reflection of the athlete's poor eccentric performance and reliance on concentric force production to maximize their jump height.

**Peak Landing Force:** High landing forces do not always indicate increased muscle strength. In sub-optimal traces, they can reflect a stiff landing strategy present in injured individuals lacking eccentric control or overall strength.

This strategy minimizes stress on muscles by avoiding energy dissipation through controlled eccentric triple flexion, leading to higher impact forces. While stiffness strategies can enhance performance in sports, athletes who lack the capacity for eccentric force attenuation may be constrained to stiff landings, limiting their movement options.

**Expert Tip:**

A shallow unloading trace and long acceleration phase often stem from low eccentric peak velocity indicating slow descent.

In healthy individuals, this may reflect poor deceleration capacity, lack of motivation, pacing or discomfort. In rehabilitation, low eccentric peak velocity may signal deceleration deficits, discomfort, pain or lack of confidence as individuals avoid the demands of a faster descent they are uncertain about controlling.



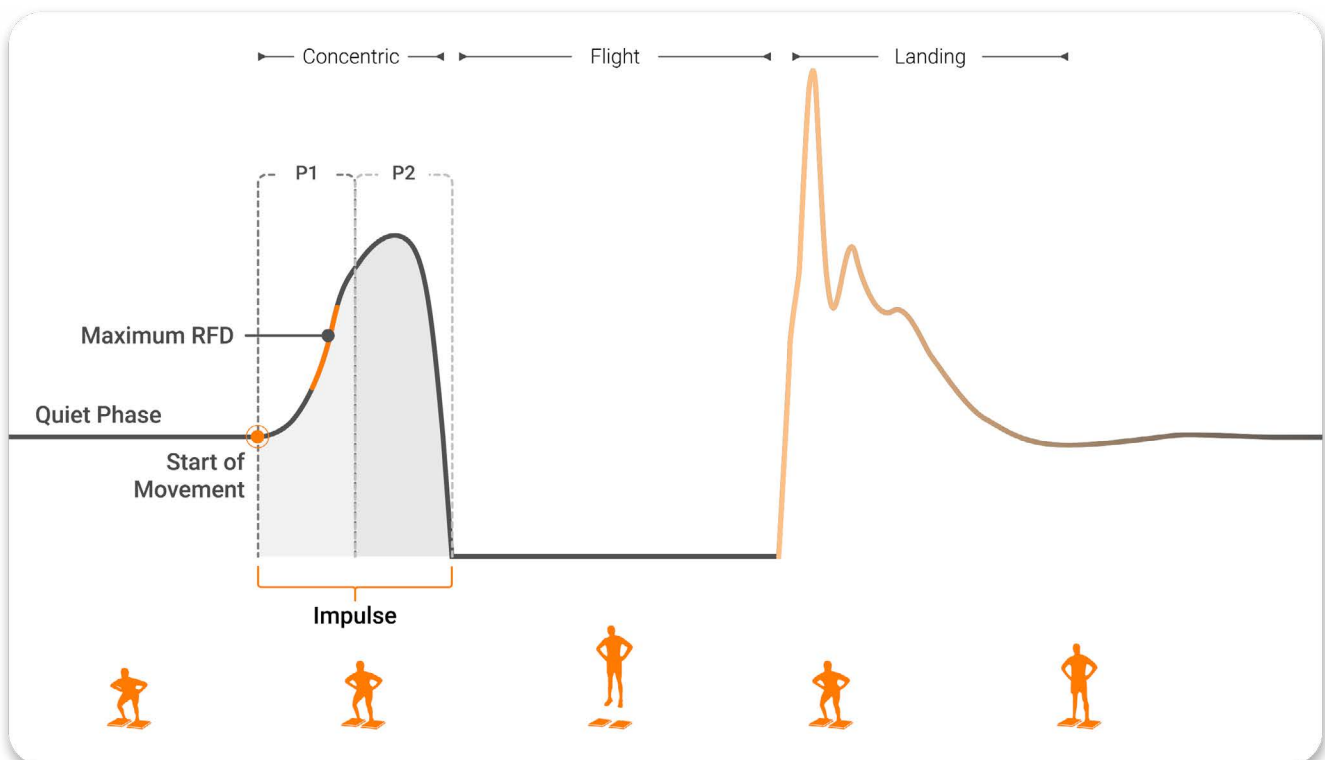
**Daniel Cohen, PhD**  
Co-Founder of ForceDecks

# Squat Jump

The SJ is a commonly executed test of concentric power and explosive strength. While similar to the CMJ, the starting position and concentric-only muscle action are key differences. The SJ begins in the individual's squat position for a steady period typically between 2-3 seconds (this can be self-selected by the individual or at a pre-specified joint angle as decided by the practitioner) and requires ONLY a concentric contraction.

By removing the eccentric contribution, the SJ is used to isolate an athlete's capacity for concentric power, explosive force expression and RFD compared to the CMJ.

Below is a visualization of an ideal force trace executed to a level that would be acceptable to include in data analysis.



**Quiet Phase:** In the SJ, the athlete starts in a squat position. It is crucial to ensure a steady, flat force trace before initiating movement. The athlete should hold this position for a count of 2-3 seconds before jumping.

**Start of Movement:** If performed correctly, the force should rise directly from the quiet phase without any countermovement. Most athletes and patients will naturally want to perform some degree of countermovement or small dip before jumping, but the aim is to cue to avoid this and to try to ensure the jump is purely concentric.

**Average RFD:** RFD is essentially the slope of the force trace (force change/time change), with the steepest point representing peak RFD. A steep, vertical trace usually indicates higher RFD values. An average RFD that is calculated between the start of movement and the peak force is one of the most reliable RFD metrics, while ForceDecks can also analyze RFD at various time intervals.

**Concentric Impulse P1:** This refers to the first 50% timewise of the concentric phase's duration, providing insights into how well the athlete pushes from the bottom position. It measures force expression from a concentric contraction, unlike the CMJ, which involves a countermovement, which allows the muscles more time to get into an activated state, allowing greater force production and jump heights.

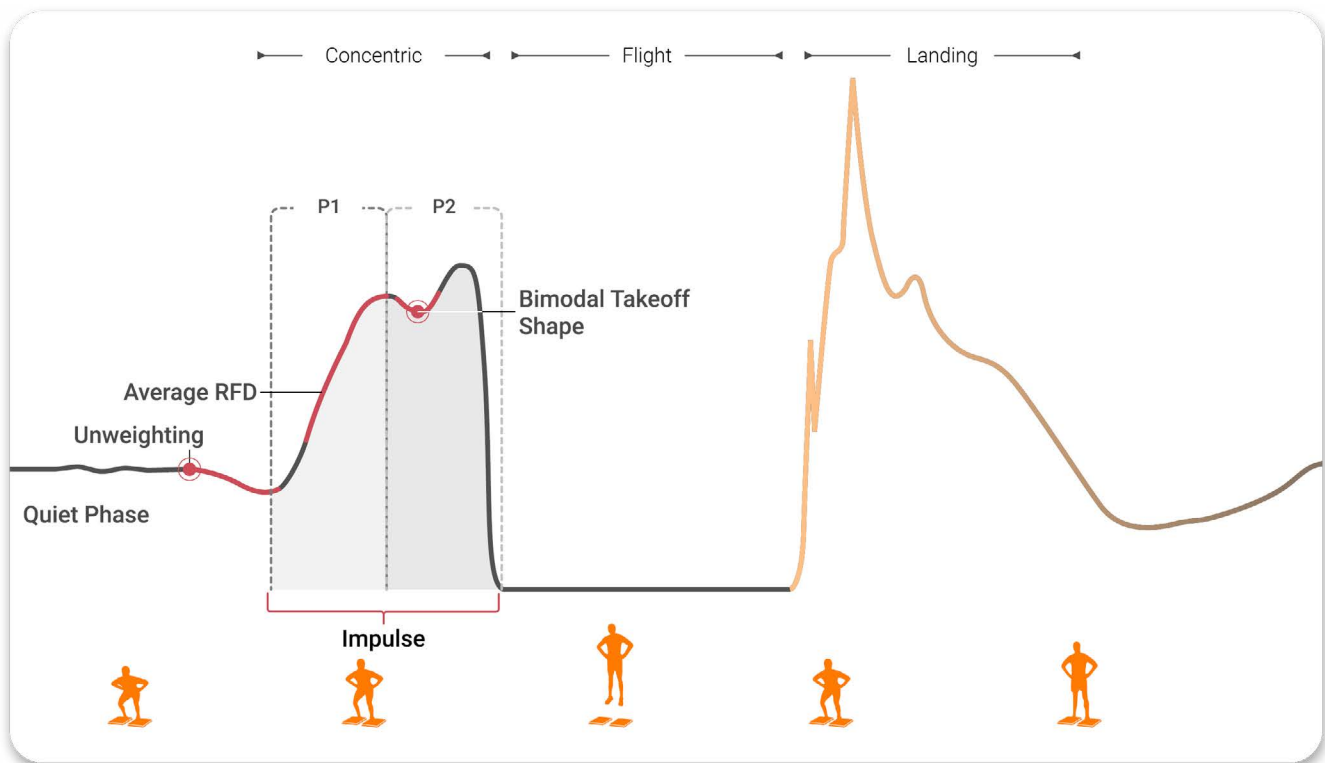
**Concentric Impulse P2 Asymmetry:** This phase focuses on concentric abilities at higher velocities and shallower joint angles as the athlete prepares for takeoff. Recent studies indicate that even when cleared to return to play, rehabbing athletes may show P2 impulse asymmetries favoring the non-injured leg.

**Flight Time:** The time between takeoff/toe-off and landing detection (force above 20N). Flight time is one of the methods used to estimate jump height and part of the FT:CT ratio. While the IMP-MOM method is often preferred for a more accurate CoM displacement, there are conditions, such as where accurate BW measurement has not been performed, that would promote the use of the flight time calculation instead.

**Landing Phase:** This shows how well the athlete decelerates upon landing, factoring in drop height, BW and landing proficiency.

However, not all force traces are created the same. It is important for performance and rehabilitation practitioners to keep their eye out for a few common force trace indicators of a sub-optimal jumping pattern.





**Quiet Phase:** The SJ will likely have a slightly less stable quiet phase than the CMJ due to the participant holding a squat position prior to movement execution. Like the CMJ, a stable pre-movement phase is important for detection of the start of the movement.

**Movement Initiation:** Whether intentional or inadvertent, participants will often perform a small countermovement before initiating the upward movement of the jump as a means of generating momentum and eliciting a small SSC action. This should be discouraged. Any repetitions with significant unweighting may need reconsideration when included in data analysis.

**Average RFD:** Average RFD in a SJ is most optimal when the curve is near vertical. However, this example of a less vertical RFD strategy may accurately reflect the individual's concentric RFD abilities, uncovering a trainable deficit to work on with the athlete.

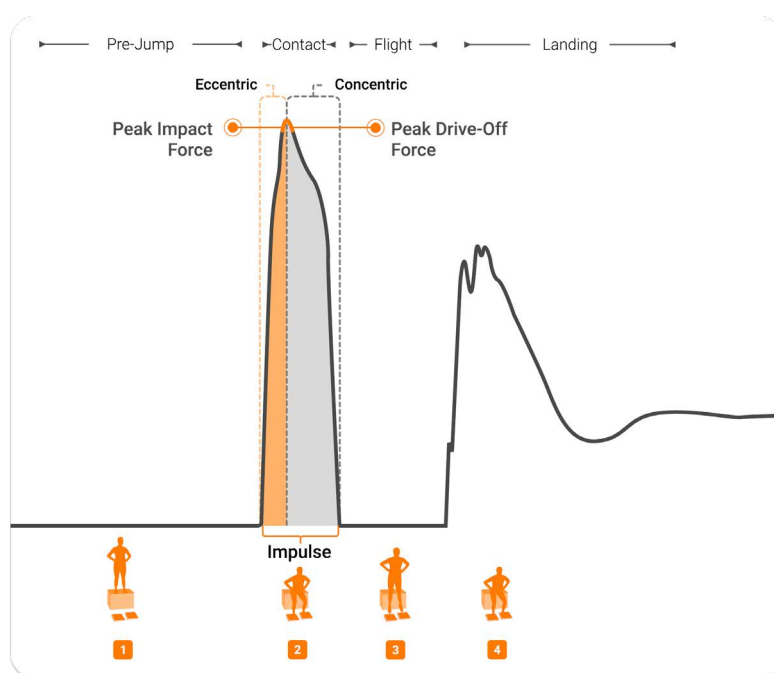
**Bimodal Takeoff:** Optimal SJ force traces feature a single smooth peak from start of movement to takeoff. However, it is common to see a bimodal or "two peak" takeoff strategy in the SJ.

SJs require a proximal to distal sequencing of muscle contractions, leading to a rapid and forceful triple extension of the lower body.

# Drop Jump

The DJ is perhaps the most utilized measure for assessing reactive strength in performance and rehabilitation settings. The DJ test involves a participant dropping from a pre-determined fixed height and quickly rebounding from the plates into a maximal jump. In completing this task, we gain insight into the athlete's ability to utilize their fast SSC.

A key point of emphasis for executing the DJ is the cue to minimize ground contact or time on the plates on landing to ensure the test captures the reactive quality it seeks. This means limiting excessive lower body flexion (especially at the hips and knees) and lowering the CoM during the contact phase to absorb the impact load before the rebound jump. This is more representative of a "depth jump" and can lead to confusion when compared to normative data.



**Quiet Phase:** While the athlete is on the box or other raised platform, the force trace should register 0N because the athlete is not on the force plates. It is important to remember that while the quiet phase registers no force, the weighing phase should still be complete prior to testing.

**Contact Time:** Total time from initial foot contact to takeoff during the jump. ForceDecks measure contact time starting when force exceeds 20N to when it drops below 20N. Contact time, essential for analyzing reactive strength, comprises the downward eccentric phase duration and upward, concentric phase duration.

**Peak Impact Force:** This is the peak force achieved upon landing on the plates from the drop. In this example, we see the force trace take the form of a steep and smooth bell curve, with a nearly symmetrical composition between the landing and drive-off portions of the contact.

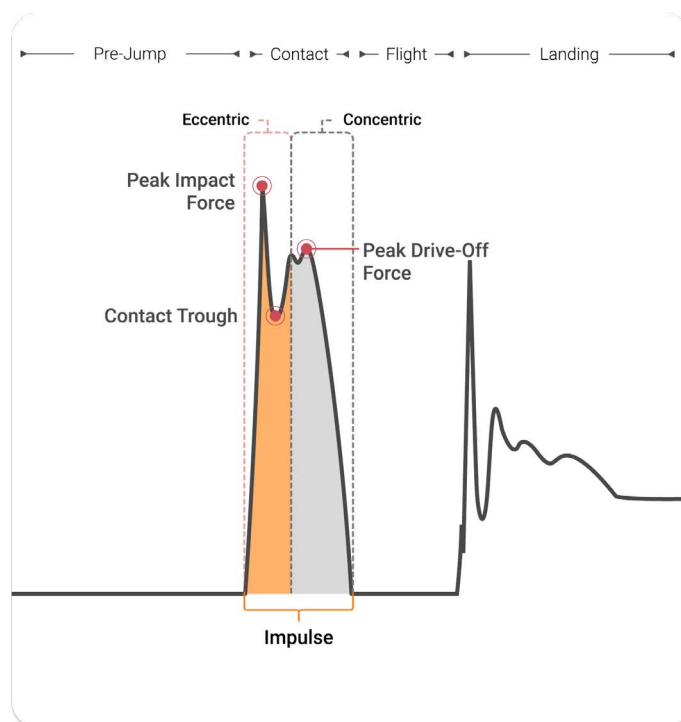
**Contact Trough:** A brief lapse in force production after peak impact force. Due to optimized utilization of the SSC, highly reactive individuals' contact trough is often minor or not visible in the force trace.

**Peak Drive-Off Force:** The greatest amount of force produced in the contact concentric phase of the jump, typically occurring in the concentric phase. Like the peak impact force, in an optimal DJ, the peak drive-off force will often be slightly higher than the peak impact force.

**Flight Time:** The time spent airborne during the jump, measured from when force drops below 20N until it exceeds 20N after landing.

**Landing Phase:** This assesses the athlete's ability to land, considering their jump height, BW and deceleration skills. Consistent landing techniques are essential for reliable metrics and should be cued for consistency.

Below is an example of a test execution more representative of a sub-optimal DJ execution.



**Contact Time:** A common symptom of a sub-optimal DJ is that individuals will have a significantly longer contact time, making other sub-optimal aspects of the takeoff phase more pronounced.

**Peak Impact Force:** This is the peak force observed upon impacting the plates from the drop. This example shows a steep impact force trace that is higher than the active drive-off force – a kinetic profile reflecting sub-optimal SSC function. The high-impact peak may also reflect a stiff quad-dominant landing strategy that does not maximize the SSC, leading to a lack of hip, knee and ankle coordination.

**Contact Trough:** In a suboptimal DJ, as a consequence of the higher impact on landing, the contact trough between this and peak drive-off is markedly more visible. Contact troughs occur for many reasons, though they are commonly observed in examples of poor utilization of the SSC.

**Peak Drive-Off Force:** The highest instantaneous force generated during the contact phase of the movement.

**Flight Time:** This is the time the athlete spends in the air during the “jump” element of the DJ. Flight time starts when force drops below 20N and continues until force returns above 20N, which marks the athlete's landing.

**Landing Phase:** Displays the athlete's ability to land after the jump. This will factor in drop height (the height they achieved in their jump), BW and their ability to decelerate proficiently. Landing strategy should be cued by the provider to ensure a consistent strategy for analysis.

# Isometric Tests

Isometric tests, while varying widely in specific test type and position, typically produce similar outcomes regarding force expression and characteristics of their force traces, although magnitudes are several-fold higher in multi-joint tests compared to isolated joint tests. Isometric tests can be unilateral, such as the RSISA series, ASH tests or bilateral, such as the IMTP or Iso Squat.

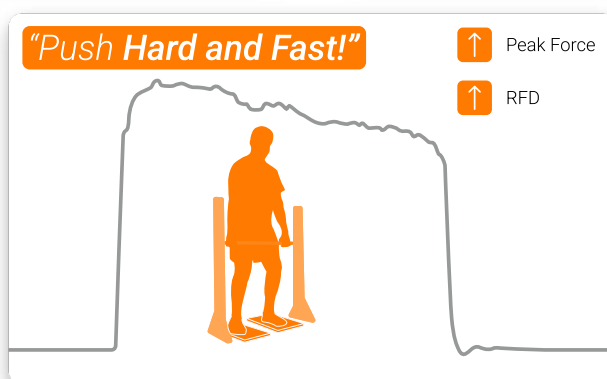
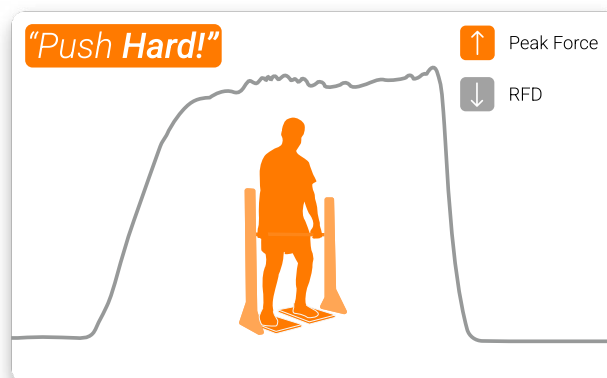
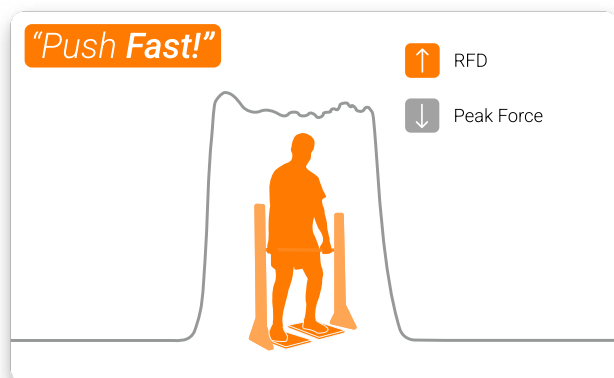
Isometric testing offers a safe solution to gain insights into an individual's capacity for maximal force production, reducing any potential risk involved in traditional strength assessments, such as a 1RM back squat or deadlift.

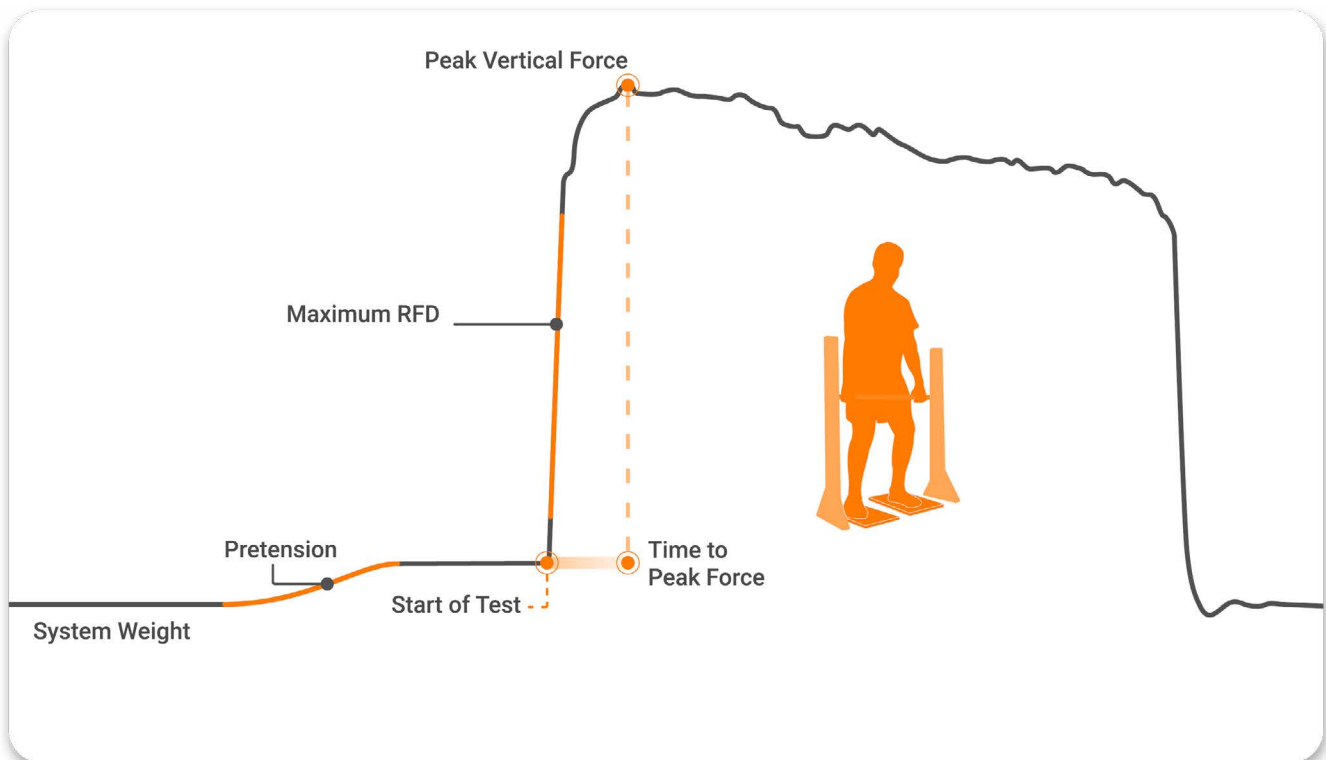
Below is a visualization of a force trace that would be acceptable to include in data analysis.



## Blind Spot: Cueing the IMTP

Although the IMTP is a relatively simple task, cueing for this test can be difficult. Cueing novice individuals to “pull” can lead to excessive upper body activity and lower overall registered forces. However, “push” may not resonate with individuals either. This is a prime example of a test that requires experience and individualized coaching to master.





**System Weight:** Equivalent to BW in standing bilateral tests, it represents the load when the athlete is resting on the force plates. In unilateral tests, like the ASH or Run-Specific Hip Iso-Push, it measures the passive, resting load (usually only a portion of BW) on the force plates just prior to the pretension phase. It is important to confirm if this weight includes additional loads (e.g., a barbell) in makeshift testing setups.

**Pretension:** Essential for both data integrity and athlete safety, pretensioning involves holding a steady tension for 3-5 seconds before initiating the movement, similar to the quiet phase in jump tests. Practitioners should aim for 5% of BW as an acceptable magnitude of pretension. This increases the likelihood of capturing metrics such as RFD accurately.

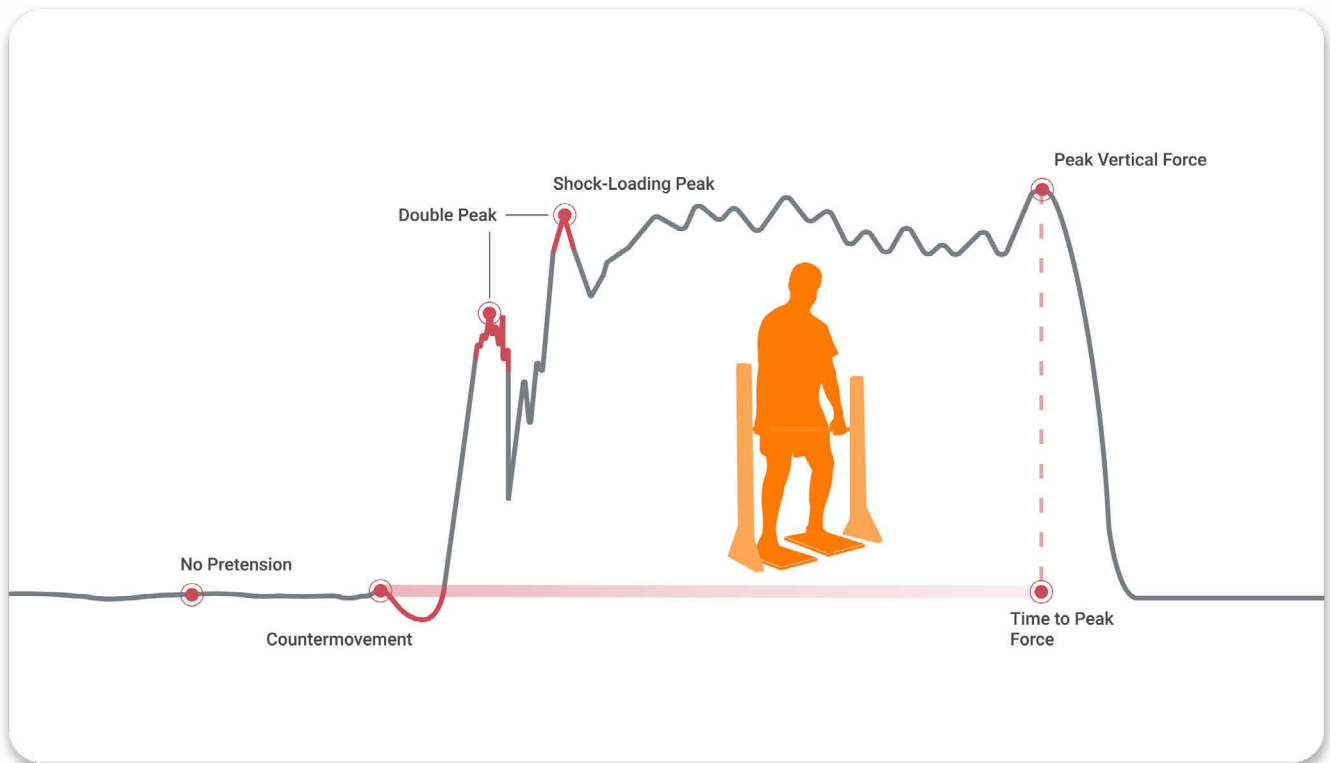
**Start of Movement:** When pretensioning is performed correctly, the start of movement occurs when the athlete produces forces greater than 20N above the pretension phase. As with the SJ test, it is crucial to avoid any countermovement to maintain data quality.

**RFD:** RFD is visible as the slope of the force-time curve, with the steepest part within a 5ms window indicating the peak or maximum RFD. RFD can be broken into both early and late phases, where early RFD (0-100ms) is often attributed to nervous system activation and late RFD (100-200ms) is a better representation of muscular-derived RFD. A tall, near-vertical force trace generally indicates higher RFD values.

**Time to Peak Force:** Reflects the athlete's strategy, which can be influenced by cues. If RFD is emphasized, time to peak force may be shorter than if maximal force is the focus.

**Peak Vertical Force:** The highest point on the force trace represents the maximum force generated during the isometric test.

Below is an example of a sub-optimal IMTP test execution, highlighting common errors.



**Pretension:** A lack of pretension in an isometric assessment is a common fault that often leads participants to execute isometric assessments improperly. Pretension is a helpful cue for the individual to create a steady low and “take out the slack” in the system prior to the maximal effort contraction, which reduces the likelihood of false spikes or peaks that can affect the accuracy of RFD values. Without it, participants might perform a countermovement to increase their momentum into the bar or force plate.

**Countermovement:** Countermovements should be discouraged because any such movement will create a pre-stretch leading into the maximal effort, which can create a falsely high reading of isometric RFD and affect start of effort detection. Isometric assessments should ideally involve no net change in joint angles during the test.

**Double Peak:** This force trace reading is another common fault resulting from implementing a momentum strategy to increase peak vertical force and RFD. This creates a “shock-loading” phenomenon where falsely high peak forces and RFD are recorded, invalidating the results.

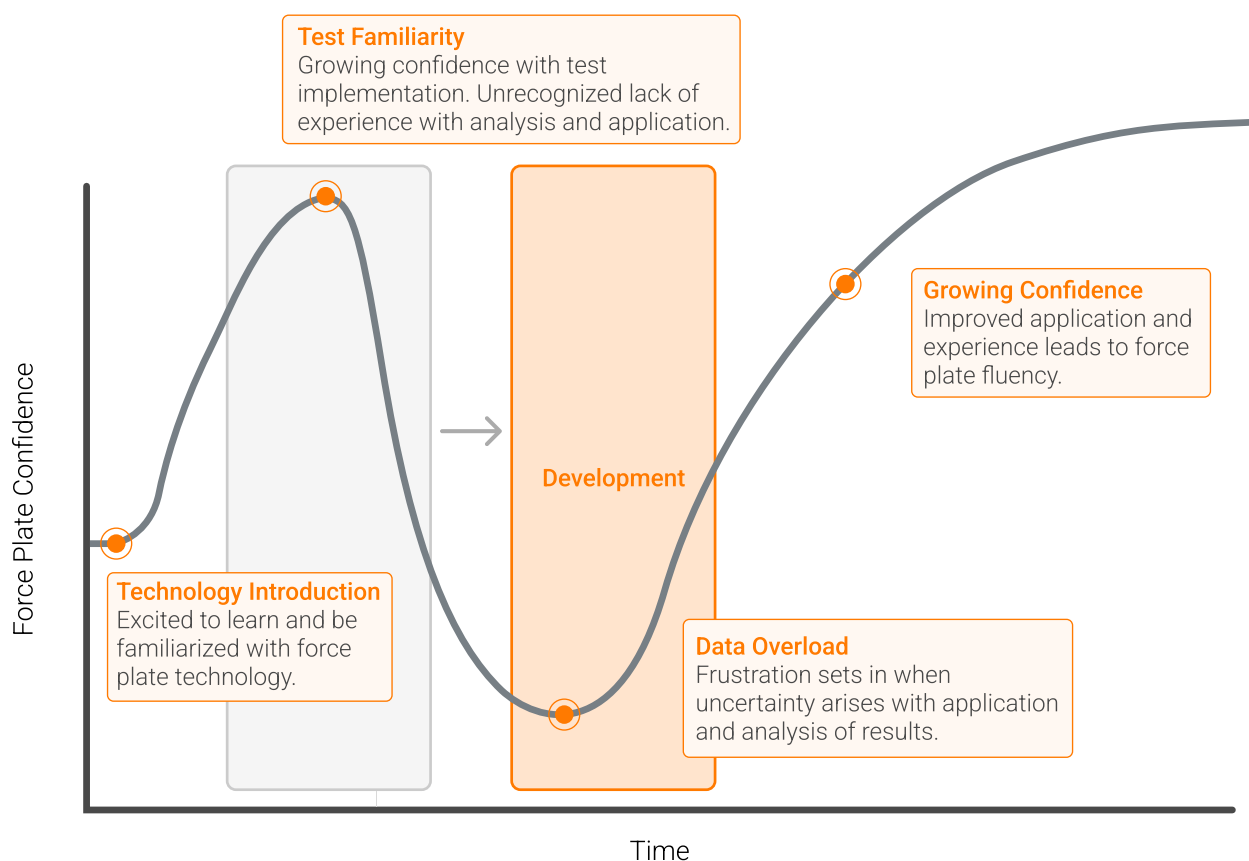
**Time to Peak Force:** This force trace has a significantly longer time to peak force. Though not inherently bad, this may provide a poor or inadequate representation of an athlete’s explosive strength capabilities.

# What Next?

This guide offers key insights into creating an assessment hierarchy, selecting force plate assessments, understanding metric utility and mastering force trace analysis. It provides a structured approach for practitioners to apply intermediate force plate concepts effectively.

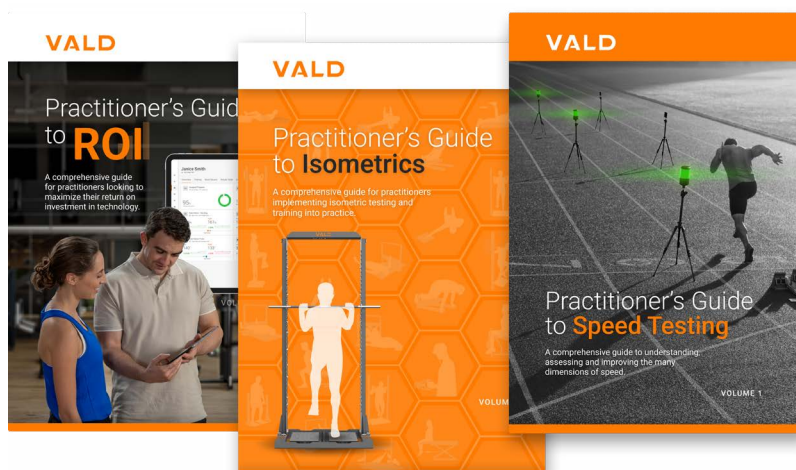
Coaches, clinicians and practitioners all come from diverse backgrounds and levels of experience. For some, this guide presents new and potentially challenging information to grasp, while others may feel it enhances their confidence with force plate use.

However, by applying the concepts outlined in this guide, all readers can develop a stronger understanding and feel more equipped to integrate force plate assessments into their practice.



# Want **More?**

VALD's [Practitioners' Series](#) offers in-depth insights for performance and health practitioners to improve their knowledge in technology-assisted training, testing and integration of modern technologies into practice. For more reading and information, check out the VALD [Practitioner's Guide to Speed](#), [Isometrics](#) and [ROI](#).



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## Practitioners Worth Following

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### Matt Jordan

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Jordan Strength



### Ryan Mc Laughlin

Sport Scientist, VALD BDM



### Paul Read

Associate Professor & S&C Coach, Return to Play  
Consultant



### Daniel Cohen

Co-Founder of ForceDecks



# Appendices

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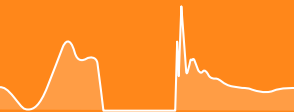
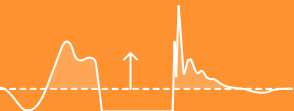
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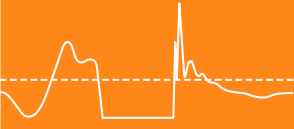
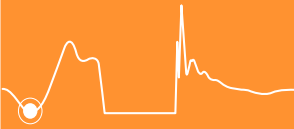
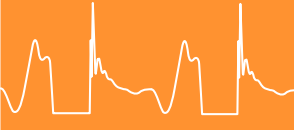
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|                                                                                     | Technical Term              | Description                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------|
|    | <b>Absolute</b>             | The magnitude of a given value (irrespective of being positive or negative).                      |
|    | <b>Acceleration</b>         | Rate of change in velocity over time of the center of mass ( $m/s^2$ ).                           |
|    | <b>Asymmetry</b>            | Relative measurement of difference between two values (R and L) expressed as a percentage.        |
|    | <b>Body Mass (BM)</b>       | Measure of total matter in an object (kg).                                                        |
|    | <b>Body Weight (BW)</b>     | Force exerted by gravity on an object's mass, captured during weighing (N).                       |
|  | <b>Center of Mass (COM)</b> | The balance point of the body that mass is distributed about.                                     |
|  | <b>Displacement</b>         | Change in the vertical position of the CoM.                                                       |
|  | <b>Force</b>                | An interaction that causes an object's motion (acceleration), deformation or both to change.      |
|  | <b>Absolute Impulse</b>     | The product of total force and time over a given or known time period.                            |
|  | <b>Net Impulse</b>          | The product of net force (total force — body weight) and time over a known time period.           |
|  | <b>Positive Impulse</b>     | The product of net force and time during a concentric action.                                     |
|  | <b>Left/Right Force</b>     | The vertical force component measured on the respective force plate on a dual force plate system. |

|                                                                                     | Technical Term           | Description                                                                                     |
|-------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------|
|    | <b>Left/Right Metric</b> | A specific quality measured on the respective force plate on a dual force plate system.         |
|    | <b>Maximum/Peak</b>      | The largest magnitude of a specific variable within a dataset or phase.                         |
|    | <b>Mean</b>              | The average of all data points in a dataset or phase, including the start and end of the phase. |
|    | <b>Minimum</b>           | The lowest magnitude of a specific variable within a dataset or phase.                          |
|   | <b>Power</b>             | The work rate a force exerts on an object, calculated as force x velocity (W).                  |
|  | <b>Recording</b>         | An entire force-time dataset from "Start Recording" to "Stop Recording" button selection.       |
|  | <b>Rep</b>               | A period in the recording where a single instance of a test or exercise is performed.           |
|  | <b>RFD</b>               | Change in force over a given time period or phase (N/s).                                        |
|  | <b>RPD</b>               | Change in power over a given time period or phase (W/s).                                        |
|  | <b>Velocity</b>          | Change in CoM position over time (m/s).                                                         |
|  | <b>Vertical Force</b>    | The force exerted perpendicular to the single force plate or combined from two.                 |

### American Football



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 30.7 cm     | 47.6 W/kg       | 0.41    |
| 25th Percentile | 36.1 cm     | 53.6 W/kg       | 0.51    |
| 50th Percentile | 41.9 cm     | 60.6 W/kg       | 0.62    |
| 75th Percentile | 47.4 cm     | 67.3 W/kg       | 0.74    |
| 90th Percentile | 52.3 cm     | 73.3 W/kg       | 0.85    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 2332 N              | 25 N/kg                  | 1450 N/s     | 1591 N/s     |
| 25th Percentile | 2781 N              | 28 N/kg                  | 2860 N/s     | 2895 N/s     |
| 50th Percentile | 3334 N              | 32.2 N/kg                | 4765 N/s     | 4592 N/s     |
| 75th Percentile | 3885 N              | 36.6 N/kg                | 6635 N/s     | 6188 N/s     |
| 90th Percentile | 4500 N              | 41.1 N/kg                | 8452 N/s     | 7782 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 27.5 cm     | 105.2 W/kg      | 0.56 s       | 1.0                                 |
| 25th Percentile | 36.3 cm     | 126.3 W/kg      | 0.47 s       | 1.4                                 |
| 50th Percentile | 44.9 cm     | 149.4 W/kg      | 0.35 s       | 1.9                                 |
| 75th Percentile | 53.3 cm     | 176.1 W/kg      | 0.27 s       | 2.5                                 |
| 90th Percentile | 60.2 cm     | 201.5 W/kg      | 0.23 s       | 3.0                                 |

### Women's Basketball



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 21.9 cm     | 38.1 W/kg       | 0.30    |
| 25th Percentile | 24.9 cm     | 41.6 W/kg       | 0.35    |
| 50th Percentile | 28.3 cm     | 45.7 W/kg       | 0.42    |
| 75th Percentile | 32.1 cm     | 50.5 W/kg       | 0.51    |
| 90th Percentile | 35.8 cm     | 54.8 W/kg       | 0.58    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1423 N              | 21.4 N/kg                | 1030 N/s     | 1109 N/s     |
| 25th Percentile | 1614 N              | 23.8 N/kg                | 1548 N/s     | 1576 N/s     |
| 50th Percentile | 1888 N              | 27.0 N/kg                | 2454 N/s     | 2402 N/s     |
| 75th Percentile | 2376 N              | 31.9 N/kg                | 3463 N/s     | 3288 N/s     |
| 90th Percentile | 2882 N              | 38.5 N/kg                | 4448 N/s     | 4249 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 16.9 cm     | 85.4 W/kg       | 0.58 s       | 1.0                                 |
| 25th Percentile | 21.5 cm     | 100.6 W/kg      | 0.47 s       | 1.1                                 |
| 50th Percentile | 26.4 cm     | 125.2 W/kg      | 0.37 s       | 1.5                                 |
| 75th Percentile | 31.2 cm     | 154.0 W/kg      | 0.28 s       | 2.0                                 |
| 90th Percentile | 36.3 cm     | 180.4 W/kg      | 0.22 s       | 2.5                                 |

### Men's Basketball



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 29.8 cm     | 46.8 W/kg       | 0.40    |
| 25th Percentile | 34.6 cm     | 52.3 W/kg       | 0.49    |
| 50th Percentile | 39.4 cm     | 57.8 W/kg       | 0.59    |
| 75th Percentile | 44.0 cm     | 63.5 W/kg       | 0.69    |
| 90th Percentile | 46.2 cm     | 68.9 W/kg       | 0.78    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1808 N              | 25.1 N/kg                | 1575 N/s     | 1695 N/s     |
| 25th Percentile | 2144 N              | 27.8 N/kg                | 2502 N/s     | 2560 N/s     |
| 50th Percentile | 2554 N              | 30.6 N/kg                | 3778 N/s     | 3640 N/s     |
| 75th Percentile | 3068 N              | 34.6 N/kg                | 5185 N/s     | 4872 N/s     |
| 90th Percentile | 3624 N              | 39.8 N/kg                | 6553 N/s     | 6104 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 24.5 cm     | 103.5 W/kg      | 0.57 s       | 1.1                                 |
| 25th Percentile | 30.5 cm     | 18.6 W/kg       | 0.48 s       | 1.4                                 |
| 50th Percentile | 38.4 cm     | 138.6 W/kg      | 0.38 s       | 1.8                                 |
| 75th Percentile | 46.4 cm     | 161.7 W/kg      | 0.29 s       | 2.2                                 |
| 90th Percentile | 54.8 cm     | 184.2 W/kg      | 0.23 s       | 2.6                                 |



DJ

### Men's Soccer



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 29.7 cm     | 45.6 W/kg       | 0.39    |
| 25th Percentile | 33.6 cm     | 49.7 W/kg       | 0.56    |
| 50th Percentile | 37.7 cm     | 54.3 W/kg       | 0.55    |
| 75th Percentile | 41.8 cm     | 59.3 W/kg       | 0.63    |
| 90th Percentile | 46.2 cm     | 64.8 W/kg       | 0.72    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1695 N              | 25.7 N/kg                | 1450 N/s     | 1552 N/s     |
| 25th Percentile | 2024 N              | 28.3 N/kg                | 2300 N/s     | 2360 N/s     |
| 50th Percentile | 2381 N              | 31.6 N/kg                | 3484 N/s     | 3406 N/s     |
| 75th Percentile | 2807 N              | 36.1 N/kg                | 4863 N/s     | 4608 N/s     |
| 90th Percentile | 3321 N              | 41.7 N/kg                | 6283 N/s     | 5756 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 23.1 cm     | 108.2 W/kg      | 0.47 s       | 1.4                                 |
| 25th Percentile | 28.5 cm     | 133.5 W/kg      | 0.33 s       | 2.0                                 |
| 50th Percentile | 34.3 cm     | 164.9 W/kg      | 0.24 s       | 2.6                                 |
| 75th Percentile | 40.0 cm     | 190.1 W/kg      | 0.19 s       | 3.0                                 |
| 90th Percentile | 45.4 cm     | 212.0 W/kg      | 0.17 s       | 3.4                                 |



DJ

### Women's Soccer



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 22.8 cm     | 38.6 W/kg       | 0.31    |
| 25th Percentile | 25.6 cm     | 41.8 W/kg       | 0.36    |
| 50th Percentile | 28.8 cm     | 45.5 W/kg       | 0.43    |
| 75th Percentile | 32.4 cm     | 49.7 W/kg       | 0.51    |
| 90th Percentile | 36.5 cm     | 54.7 W/kg       | 0.59    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1357 N              | 22.3 N/kg                | 1190 N/s     | 1256 N/s     |
| 25th Percentile | 1563 N              | 24.7 N/kg                | 1740 N/s     | 1772 N/s     |
| 50th Percentile | 1832 N              | 28.3 N/kg                | 2493 N/s     | 2436 N/s     |
| 75th Percentile | 2154 N              | 32.8 N/kg                | 3339 N/s     | 3212 N/s     |
| 90th Percentile | 2528 N              | 38.1 N/kg                | 4255 N/s     | 3985 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 16.2 cm     | 85.4 W/kg       | 0.55 s       | 1.0                                 |
| 25th Percentile | 21.8 cm     | 100.6 W/kg      | 0.42 s       | 1.3                                 |
| 50th Percentile | 27.2 cm     | 125.2 W/kg      | 0.30 s       | 1.8                                 |
| 75th Percentile | 32.0 cm     | 154.0 W/kg      | 0.22 s       | 2.4                                 |
| 90th Percentile | 36.2 cm     | 180.4 W/kg      | 0.19 s       | 2.9                                 |

### Women's AFL



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 22.9 cm     | 38.8 W/kg       | 0.30    |
| 25th Percentile | 26.0 cm     | 42.0 W/kg       | 0.37    |
| 50th Percentile | 29.2 cm     | 45.9 W/kg       | 0.43    |
| 75th Percentile | 32.6 cm     | 50.2 W/kg       | 0.51    |
| 90th Percentile | 38.1 cm     | 55.8 W/kg       | 0.60    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1520 N              | 22.0 N/kg                | 1198 N/s     | 1331 N/s     |
| 25th Percentile | 1791 N              | 25.7 N/kg                | 1828 N/s     | 1931 N/s     |
| 50th Percentile | 2128 N              | 31.2 N/kg                | 2965 N/s     | 2982 N/s     |
| 75th Percentile | 2557 N              | 36.3 N/kg                | 4521 N/s     | 4221 N/s     |
| 90th Percentile | 2968 N              | 39.9 N/kg                | 6116 N/s     | 5631 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 19.6 cm     | 92.8 W/kg       | 0.46 s       | 1.2                                 |
| 25th Percentile | 24.3 cm     | 107.5 W/kg      | 0.37 s       | 1.6                                 |
| 50th Percentile | 28.8 cm     | 134.8 W/kg      | 0.27 s       | 2.1                                 |
| 75th Percentile | 33.1 cm     | 159.0 W/kg      | 0.22 s       | 2.6                                 |
| 90th Percentile | 36.6 cm     | 183.1 W/kg      | 0.19 s       | 3.0                                 |



DJ

### Men's AFL



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 29.9 cm     | 46.0 W/kg       | 0.40    |
| 25th Percentile | 33.6 cm     | 50.4 W/kg       | 0.48    |
| 50th Percentile | 37.4 cm     | 55.3 W/kg       | 0.58    |
| 75th Percentile | 41.8 cm     | 60.9 W/kg       | 0.68    |
| 90th Percentile | 47.6 cm     | 68.1 W/kg       | 0.80    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 2080 N              | 27.8 N/kg                | 1288 N/s     | 1466 N/s     |
| 25th Percentile | 2481 N              | 31.3 N/kg                | 2404 N/s     | 2556 N/s     |
| 50th Percentile | 2900 N              | 35.4 N/kg                | 3955 N/s     | 3945 N/s     |
| 75th Percentile | 3348 N              | 40.0 N/kg                | 5689 N/s     | 5344 N/s     |
| 90th Percentile | 3775 N              | 44.2 N/kg                | 7300 N/s     | 6636 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 22.2 cm     | 126.0 W/kg      | 0.34 s       | 1.8                                 |
| 25th Percentile | 28.7 cm     | 150.2 W/kg      | 0.26 s       | 2.2                                 |
| 50th Percentile | 34.2 cm     | 170.8 W/kg      | 0.22 s       | 2.6                                 |
| 75th Percentile | 39.4 cm     | 193.1 W/kg      | 0.19 s       | 3.0                                 |
| 90th Percentile | 44.2 cm     | 218.3 W/kg      | 0.17 s       | 3.4                                 |

## Women's Track & Field



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 22.6 cm     | 38.0 W/kg       | 0.31    |
| 25th Percentile | 26.3 cm     | 43.5 W/kg       | 0.38    |
| 50th Percentile | 31.9 cm     | 50.6 W/kg       | 0.49    |
| 75th Percentile | 37.0 cm     | 56.9 W/kg       | 0.61    |
| 90th Percentile | 41.7 cm     | 63.1 W/kg       | 0.73    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1320 N              | 22.8 N/kg                | 1195 N/s     | 1225 N/s     |
| 25th Percentile | 1596 N              | 25.5 N/kg                | 1946 N/s     | 1952 N/s     |
| 50th Percentile | 1962 N              | 29.9 N/kg                | 2836 N/s     | 2665 N/s     |
| 75th Percentile | 2404 N              | 35.0 N/kg                | 3711 N/s     | 3521 N/s     |
| 90th Percentile | 3120 N              | 45.8 N/kg                | 4785 N/s     | 4438 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 21.0 cm     | 104.0 W/kg      | 0.43 s       | 1.4                                 |
| 25th Percentile | 26.5 cm     | 132.8 W/kg      | 0.32 s       | 1.8                                 |
| 50th Percentile | 31.3 cm     | 167.4 W/kg      | 0.23 s       | 2.6                                 |
| 75th Percentile | 36.7 cm     | 201.5 W/kg      | 0.19 s       | 3.2                                 |
| 90th Percentile | 43.6 cm     | 225.2 W/kg      | 0.16 s       | 3.6                                 |



DJ

### Men's Rugby



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 29.6 cm     | 46.9 W/kg       | 0.38    |
| 25th Percentile | 34.5 cm     | 51.7 W/kg       | 0.48    |
| 50th Percentile | 39.5 cm     | 56.9 W/kg       | 0.58    |
| 75th Percentile | 44.6 cm     | 62.3 W/kg       | 0.67    |
| 90th Percentile | 49.3 cm     | 67.7 W/kg       | 0.75    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 2105 N              | 27.6 N/kg                | 1350 N/s     | 1476 N/s     |
| 25th Percentile | 2619 N              | 31.1 N/kg                | 2441 N/s     | 2623 N/s     |
| 50th Percentile | 3210 N              | 35.5 N/kg                | 4060 N/s     | 4112 N/s     |
| 75th Percentile | 3810 N              | 40.2 N/kg                | 5990 N/s     | 5842 N/s     |
| 90th Percentile | 4329 N              | 45.1 N/kg                | 7866 N/s     | 7219 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 24.3 cm     | 117.3 W/kg      | 0.40 s       | 1.6                                 |
| 25th Percentile | 29.7 cm     | 150.3 W/kg      | 0.28 s       | 2.2                                 |
| 50th Percentile | 34.5 cm     | 176.2 W/kg      | 0.22 s       | 2.8                                 |
| 75th Percentile | 39.2 cm     | 195.8 W/kg      | 0.19 s       | 3.1                                 |
| 90th Percentile | 44.4 cm     | 211.6 W/kg      | 0.18 s       | 3.4                                 |

### Men's Track & Field



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 27.9 cm     | 43.9 W/kg       | 0.37    |
| 25th Percentile | 33.1 cm     | 50.8 W/kg       | 0.48    |
| 50th Percentile | 41.6 cm     | 60.0 W/kg       | 0.62    |
| 75th Percentile | 47.9 cm     | 67.7 W/kg       | 0.76    |
| 90th Percentile | 53.5 cm     | 75.4 W/kg       | 0.91    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1692 N              | 25.8 N/kg                | 1440 N/s     | 1536 N/s     |
| 25th Percentile | 2135 N              | 29.6 N/kg                | 2454 N/s     | 2535 N/s     |
| 50th Percentile | 2749 N              | 34.8 N/kg                | 4028 N/s     | 3817 N/s     |
| 75th Percentile | 3689 N              | 44.2 N/kg                | 6002 N/s     | 5709 N/s     |
| 90th Percentile | 4682 N              | 55.5 N/kg                | 8333 N/s     | 7720 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 23.4 cm     | 129.1 W/kg      | 0.37 s       | 1.8                                 |
| 25th Percentile | 31.2 cm     | 155.1 W/kg      | 0.30 s       | 2.2                                 |
| 50th Percentile | 40.1 cm     | 188.7 W/kg      | 0.24 s       | 2.6                                 |
| 75th Percentile | 45.7 cm     | 217.7 W/kg      | 0.20 s       | 3.2                                 |
| 90th Percentile | 52.4 cm     | 247.0 W/kg      | 0.17 s       | 3.6                                 |



DJ

### Men's Rugby



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 29.6 cm     | 46.9 W/kg       | 0.38    |
| 25th Percentile | 34.5 cm     | 51.7 W/kg       | 0.48    |
| 50th Percentile | 39.5 cm     | 56.9 W/kg       | 0.58    |
| 75th Percentile | 44.6 cm     | 62.3 W/kg       | 0.67    |
| 90th Percentile | 49.3 cm     | 67.7 W/kg       | 0.75    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 2105 N              | 27.6 N/kg                | 1350 N/s     | 1476 N/s     |
| 25th Percentile | 2619 N              | 31.1 N/kg                | 2441 N/s     | 2623 N/s     |
| 50th Percentile | 3210 N              | 35.5 N/kg                | 4060 N/s     | 4112 N/s     |
| 75th Percentile | 3810 N              | 40.2 N/kg                | 5990 N/s     | 5842 N/s     |
| 90th Percentile | 4329 N              | 45.1 N/kg                | 7866 N/s     | 7219 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 24.3 cm     | 117.3 W/kg      | 0.40 s       | 1.6                                 |
| 25th Percentile | 29.7 cm     | 150.3 W/kg      | 0.28 s       | 2.2                                 |
| 50th Percentile | 34.5 cm     | 176.2 W/kg      | 0.22 s       | 2.8                                 |
| 75th Percentile | 39.2 cm     | 195.8 W/kg      | 0.19 s       | 3.1                                 |
| 90th Percentile | 44.4 cm     | 211.6 W/kg      | 0.18 s       | 3.4                                 |

### Women's Rugby



CMJ

|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 22.9 cm     | 37.9 W/kg       | 0.34    |
| 25th Percentile | 26.1 cm     | 42.0 W/kg       | 0.41    |
| 50th Percentile | 29.8 cm     | 46.3 W/kg       | 0.49    |
| 75th Percentile | 33.8 cm     | 51.1 W/kg       | 0.57    |
| 90th Percentile | 37.9 cm     | 56.0 W/kg       | 0.66    |



IMTP

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1486 N              | 23.7 N/kg                | 1150 N/s     | 1288 N/s     |
| 25th Percentile | 1790 N              | 26.5 N/kg                | 1805 N/s     | 2023 N/s     |
| 50th Percentile | 2087 N              | 30.4 N/kg                | 2820 N/s     | 2840 N/s     |
| 75th Percentile | 2505 N              | 34.3 N/kg                | 3895 N/s     | 3856 N/s     |
| 90th Percentile | 2025 N              | 37.8 N/kg                | 5130 N/s     | 4820 N/s     |



DJ

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 14.5 cm     | 80.3 W/kg       | 0.48 s       | 1.1                                 |
| 25th Percentile | 22.1 cm     | 96.5 W/kg       | 0.42 s       | 1.3                                 |
| 50th Percentile | 26.8 cm     | 117.1 W/kg      | 0.34 s       | 1.7                                 |
| 75th Percentile | 31.7 cm     | 143.2 W/kg      | 0.26 s       | 2.1                                 |
| 90th Percentile | 35.7 cm     | 167.4 W/kg      | 0.21 s       | 2.5                                 |

### Men's Ice Hockey



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 31.3 cm     | 46.0 W/kg       | 0.41    |
| 25th Percentile | 35.5 cm     | 50.2 W/kg       | 0.48    |
| 50th Percentile | 39.5 cm     | 54.7 W/kg       | 0.57    |
| 75th Percentile | 43.4 cm     | 59.3 W/kg       | 0.66    |
| 90th Percentile | 47.3 cm     | 63.3 W/kg       | 0.73    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1946 N              | 27.3 N/kg                | 1540 N/s     | 1696 N/s     |
| 25th Percentile | 2303 N              | 30.0 N/kg                | 2410 N/s     | 2580 N/s     |
| 50th Percentile | 2749 N              | 34.5 N/kg                | 3655 N/s     | 3646 N/s     |
| 75th Percentile | 3300 N              | 40.6 N/kg                | 5175 N/s     | 4935 N/s     |
| 90th Percentile | 4120 N              | 51.8 N/kg                | 6750 N/s     | 6348 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 23.3 cm     | 111.7 W/kg      | 0.59 s       | 1.1                                 |
| 25th Percentile | 23.4 cm     | 128.7 W/kg      | 0.41 s       | 1.6                                 |
| 50th Percentile | 34.8 cm     | 147.9 W/kg      | 0.25 s       | 2.3                                 |
| 75th Percentile | 40.9 cm     | 165.8 W/kg      | 0.21 s       | 2.7                                 |
| 90th Percentile | 46.7 cm     | 179.8 W/kg      | 0.19 s       | 3.1                                 |



DJ

### Women's Ice Hockey



|                 | Jump Height | Peak Power / BM | RSI-Mod |
|-----------------|-------------|-----------------|---------|
| 10th Percentile | 22.1 cm     | 36.9 W/kg       | 0.27    |
| 25th Percentile | 25.1 cm     | 40.0 W/kg       | 0.32    |
| 50th Percentile | 27.9 cm     | 43.1 W/kg       | 0.39    |
| 75th Percentile | 31.2 cm     | 46.9 W/kg       | 0.46    |
| 90th Percentile | 35.6 cm     | 51.6 W/kg       | 0.53    |



CMJ

|                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|-----------------|---------------------|--------------------------|--------------|--------------|
| 10th Percentile | 1648 N              | 25.2 N/kg                | 1257 N/s     | 1517 N/s     |
| 25th Percentile | 1924 N              | 28.1 N/kg                | 2241 N/s     | 2297 N/s     |
| 50th Percentile | 2207 N              | 31.7 N/kg                | 3278 N/s     | 3186 N/s     |
| 75th Percentile | 2504 N              | 35.4 N/kg                | 4191 N/s     | 3914 N/s     |
| 90th Percentile | 2853 N              | 39.7 N/kg                | 5341 N/s     | 4833 N/s     |



IMTP

|                 | Jump Height | Peak Power / BM | Contact Time | RSI<br>(Flight Time / Contact Time) |
|-----------------|-------------|-----------------|--------------|-------------------------------------|
| 10th Percentile | 18.5 cm     | 84.0 W/kg       | 0.60 s       | 0.8                                 |
| 25th Percentile | 23.2 cm     | 90.6 W/kg       | 0.52 s       | 1.1                                 |
| 50th Percentile | 27.7 cm     | 119.5 W/kg      | 0.34 s       | 1.8                                 |
| 75th Percentile | 32.1 cm     | 142.8 W/kg      | 0.25 s       | 2.2                                 |
| 90th Percentile | 35.0 cm     | 177.5 W/kg      | 0.20 s       | 2.7                                 |



DJ

### Men's AFL



|                                                                                          |       | 25th Percentile | 50th Percentile | 90th Percentile |
|------------------------------------------------------------------------------------------|-------|-----------------|-----------------|-----------------|
| <br>CMJ | >18   | 29.9 cm         | 46.0 W/kg       | 0.40            |
|                                                                                          | 18-25 | 33.6 cm         | 50.4 W/kg       | 0.48            |
|                                                                                          | 26-30 | 37.4 cm         | 55.3 W/kg       | 0.58            |
|                                                                                          | 31-35 | 41.8 cm         | 60.9 W/kg       | 0.68            |
|                                                                                          | 35-40 | 47.6 cm         | 68.1 W/kg       | 0.80            |
|                                                                                          | 41-45 | 53.5 cm         | 76.2 W/kg       | 0.90            |

|                                                                                             |                 | Peak Vertical Force | Peak Vertical Force / BM | RFD at 200ms | RFD at 250ms |
|---------------------------------------------------------------------------------------------|-----------------|---------------------|--------------------------|--------------|--------------|
| <br>IMTP | 10th Percentile | 2080 N              | 27.8 N/kg                | 1288 N/s     | 1466 N/s     |
|                                                                                             | 25th Percentile | 2481 N              | 31.3 N/kg                | 2404 N/s     | 2556 N/s     |
|                                                                                             | 50th Percentile | 2900 N              | 35.4 N/kg                | 3955 N/s     | 3945 N/s     |
|                                                                                             | 75th Percentile | 3348 N              | 40.0 N/kg                | 5689 N/s     | 5344 N/s     |
|                                                                                             | 90th Percentile | 3775 N              | 44.2 N/kg                | 7300 N/s     | 6636 N/s     |

|                                                                                           |                 | Jump Height | Peak Power / BM | Contact Time | RSI<br><small>(Flight Time / Contact Time)</small> |
|-------------------------------------------------------------------------------------------|-----------------|-------------|-----------------|--------------|----------------------------------------------------|
| <br>DJ | 10th Percentile | 22.2 cm     | 126.0 W/kg      | 0.34 s       | 1.8                                                |
|                                                                                           | 25th Percentile | 28.7 cm     | 150.2 W/kg      | 0.26 s       | 2.2                                                |
|                                                                                           | 50th Percentile | 34.2 cm     | 170.8 W/kg      | 0.22 s       | 2.6                                                |
|                                                                                           | 75th Percentile | 39.4 cm     | 193.1 W/kg      | 0.19 s       | 3.0                                                |
|                                                                                           | 90th Percentile | 44.2 cm     | 218.3 W/kg      | 0.17 s       | 3.4                                                |



# Get ready to accelerate your force plate competency. **Let's jump in.**

This guide provides insights into force plate technology, enabling you to optimize testing, understand key metrics and tailor assessments to your athletes' or patients' needs.

For users who have reached this intermediate stage of force plate knowledge and experience, congratulations! However, the road to force plate mastery is a long one, so beware of overconfidence in force plate testing and embrace the opportunity to continue your learning journey.

## Some questions we'll answer in this guide:

**Q:** How do I choose **which metrics** to pay attention to?

**Q:** How do I select **which tests** I should use?

**Q:** How do I ensure I am getting **accurate data** from my tests?

**Q:** How do I develop **physical characteristics** with force plates?

**Q:** How do I interpret **force-time curves**?

Want to learn more? We are here to help.

**info@vald.com**